MultiSenseSeg: A Cost-Effective Unified Multimodal Semantic Segmentation Model for Remote Sensing

Qingpeng Wang, Wei Chen, Zhou Huang, Hongzhao Tang, and Lan Yang

***Abstract*— Semantic segmentation is an essential technique in remote sensing. Until recently, most related research has focused primarily on advancing semantic segmentation models based on monomodal imagery, and less attention has been given to models that utilize multimodal remote sensing data. Moreover, most current multimodal approaches consider only limited bimodal situations and cannot simultaneously utilize three or more modalities. The increase in expensive computational costs associated with previous feature fusion paradigms hinders their application in broader cases. How to design a unified method to cover a wide variety of quantity-agnostic modalities for multimodal semantic segmentation remains an unsolved issue. To address the aforementioned challenges, this study explores a feasible way and proposes a cost-effective multimodal sensing semantic segmentation model (MultiSenseSeg). MultiSenseSeg employs multiple lightweight modality-specific experts (MSEs), an adaptive multimodal matching (AMM) module, and a single feature extraction pipeline to efficiently model intramodal and intermodal relationships. Benefiting from these designs, the proposed MultiSenseSeg can serve as a unified multimodal model capable of addressing both monomodal and bimodal cases and readily extrapolating to scenarios with more modalities, thereby achieving semantic segmentation of arbitrary quantities of multimodal data. To evaluate the performance of our method, we select several state-of-the-art (SOTA) semantic segmentation models from the past three years and conduct extensive experiments on two public multimodal datasets. The results show that MultiSenseSeg can not only achieve higher accuracy but also exhibit user-friendly modality extrapolation, allowing end-to-end training for consumer-grade users based on limited hardware resources. The model's code will be available at https://github.com/W-qp/MultiSenseSeg.**

***Index Terms*— Deep learning, low-cost increment, multimodal, remote sensing, semantic segmentation.**

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Qingpeng Wang is with the College of Geoscience and Surveying Engineering, China University of Mining and Technology, Beijing 100083, China, and also with the Institute of Remote Sensing and Geographic Information System, School of Earth and Space Sciences, Peking University, Beijing 100871, China (e-mail: wangqp0313@126.com).

Wei Chen is with the College of Geoscience and Surveying Engineering, China University of Mining and Technology, Beijing 100083, China (e-mail: chenw@cumtb.edu.cn).

Zhou Huang is with the Institute of Remote Sensing and Geographic Information System, School of Earth and Space Sciences, Peking University, Beijing 100871, China (e-mail: huangzhou@pku.edu.cn).

Hongzhao Tang is with the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources, Beijing 100048, China (e-mail: tanghz@pku.edu.cn).

Lan Yang is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, Hubei 430079, China.

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I. Introduction

SEMANTIC segmentation is one of the most advanced scene-aware methods for remote sensing; it aims to assign a semantic category to each image pixel [1], i.e., to identify and classify objects at the pixel level. It has many crucial applications in remote sensing, including building footprint extraction [2], [3], environmental preservation [4], [5], land cover mapping [6], [7], and precision agriculture [8], [9]. However, most of the aforementioned studies have focused mainly on using a single image modality as the model's input, i.e., performing monomodal semantic segmentation. For some complex scenarios, such as two object classes with similar textures and colors, monomodal models may struggle to distinguish them due to limited perceptual information. Therefore, some studies introduce extra modalities to overcome performance bottlenecks [10], [11]. For instance, combining optical imagery with digital surface model (DSM) data could help in enhancing interclass differences and obtaining more precise semantic segmentation results.

Multimodal semantic segmentation models can improve segmentation by exploiting complementary modalities [12], [13], [14]. These supplementary modalities encompass multisource data of the same modality (such as visible light, multispectral, and hyperspectral data) and data from different modalities (such as LiDAR, polarimetric, and nighttime data); for simplicity, this article hereafter collectively terms them as multimodal data. With the development of photogrammetric sensors, there is an increasingly diverse variety of data types that are highly applicable for multimodal semantic segmentation. However, compared with the rapid development of monomodal models, multimodal semantic segmentation is still in its early stages, and the advantages of remote sensing diverse data sources have not been sufficiently exploited.

Previous multimodal methods are usually designed for bimodal scenarios (which can only input two types of data) [15], [16], lacking the ability to utilize more than two modalities simultaneously. Although some existing bimodal models can be extended to multiple situations according to their paradigm, the significant computational cost increment and inefficiency incurred by extrapolating the model to quantity-agnostic modalities remain unsolved problems.

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Specifically, when extrapolating, it is inevitable that a new feature extraction backbone is allocated for each extra modality. The computational cost associated with this part increases linearly [17], [18], ultimately leading to a large increase in the GPU load. Therefore, this method is at the cost of inefficiency, making it challenging to offer practical usage for general users as the number of input modalities continues to increase. Another concern is the burden of model redesign. Users are required to manually perform secondary modifications to the dual-input architecture, which is evidently inefficient. Therefore, a challenging research question arises: how to construct a cost-effective, high-efficiency unified model to cover a wide variety of quantity-agnostic modalities for multimodal semantic segmentation?

To solve the aforementioned question, this study explored a novel scalable way to construct a unified semantic segmentation model toward unlimited remote sensing modalities. We propose MultiSenseSeg, a highly extensible multimodal sensing semantic segmentation model that can address quantity-agnostic modality tasks (including monomodal and bimodal situations), and the computational cost increment for each new extra modality is very small. Considering that the extrapolation cost grows linearly, significantly limiting the optimization space for the model, we design lightweight modality-specific experts (MSEs), an adaptive multimodal matching (AMM) module, and a feature extraction pipeline. These components relax the representation of multimodal features, allowing multimodal feature maps to transition from low to high dimensions, and then compress back into a low-dimensional space for processing instead of consistently maintaining a high-dimensional data structure at the encoding end. Moreover, from a feature extraction perspective, they efficiently model intramodal and intermodal relationships, i.e., modal gaps and modality diversity. Specifically, first, there are differences in the data distributions among modalities (e.g., the density of optical images and the sparsity of geometric points). MSEs can extract modality-specific semantic features while narrowing the large gap across modalities, preventing the direct use of homogeneous operators to generate lossy features. Second, unlike monomodal and bimodal scenarios, quantity-agnostic modalities carry more diverse semantics, so the feature matrices in the network have a higher dimensional data structure. Retaining high-order feature maps for operations is inefficient. AMM routes through high-order multimodal global space, searches for complementary information for activation, and achieves dimensionality reduction to losslessly integrate high-quality multimodal features in the form of low-rank matrices. Subsequently, a single optimized feature extraction pipeline is employed to learn multimodal features from global to local and coupled with a semantic segmentation decoder. Notably, we adopt a consistent topology design for MSEs, thereby enabling users to only tune a hyperparameter of our model to effortlessly extend it to multiple situations, eliminating the need for secondary modifications to the network structure.

We conduct extensive experiments to evaluate the performance of MultiSenseSeg. Comparative experimental results with several state-of-the-art (SOTA) models on two public multimodal datasets show that our method not only achieves higher accuracy but also incurs minimal incremental computational cost for each extra input modality. Specifically, under the same computational conditions, the parameter (Params) cost incurred by MultiSenseSeg for processing each extra modality was only 0.08 M, whereas the comparison models require at least 20 M. Furthermore, even under the extreme case of simultaneously handling 20 different input modalities, MultiSenseSeg still maintains robust performance. In summary, our main contributions are given as follows.

1) We rethink multimodal semantic segmentation from a broader perspective and explore a feasible way to construct a universal semantic segmentation model (MultiSenseSeg) when dealing with quantity-agnostic modalities. The proposed MultiSenseSeg can achieve comparable or even better accuracy than some SOTA techniques on two public multimodal datasets.
2) We design a novel lightweight multimodal backbone for MultiSenseSeg, composed of MSE, AMM, and feature extraction pipeline. Benefiting from these designs, the extrapolation of MultiSenseSeg is user-friendly and does not rely on secondary rewriting of the network architecture, and the incremental computational cost for each extra modality is minimal.

The remainder of this article is organized as follows. In Section II, we review relevant works on monomodal and bimodal semantic segmentation. In Section III, we provide a detailed exposition of the MultiSenseSeg architecture. Section IV delineates the preprocessing methodology for multimodal datasets and describes the experimental setup. In Section V, we present the benchmark experimental results. Section VI introduces the ablation study of the MultiSenseSeg. Section VII is the summary and conclusion.

II. Related Works

A. Monomodal Semantic Segmentation Methods

Monomodal semantic segmentation refers to models that can utilize only one type of data source as input. FCN [19] is the pioneering monomodal semantic segmentation model. Subsequent studies introduced various convolutional neural network (CNN) architectures, including standard encoder–decoder architectures [20], [21], skip [22], [23]/residual [24], [25]/dense [26], [27] connections, contextual dependencies [28], [29], and multiscale strategies [30], [31], [32]. These tricks have played an important role in segmentation tasks based on high-resolution remote sensing images [33], [34], [35].

Recently, as a formidable competitor to the CNN camp, vision transformers (ViTs) have swept the domain of image processing, including remote sensing [1], [36], [37]. Benefiting from flexible self-attention, ViTs possess larger effective receptive fields than CNNs, enabling them to learn more robust visual representations. The ViT [38] is one of the most representative models and has achieved great success in vision tasks owing to its global receptive fields. However, its low-resolution feature maps, global quadratic computational complexity, and

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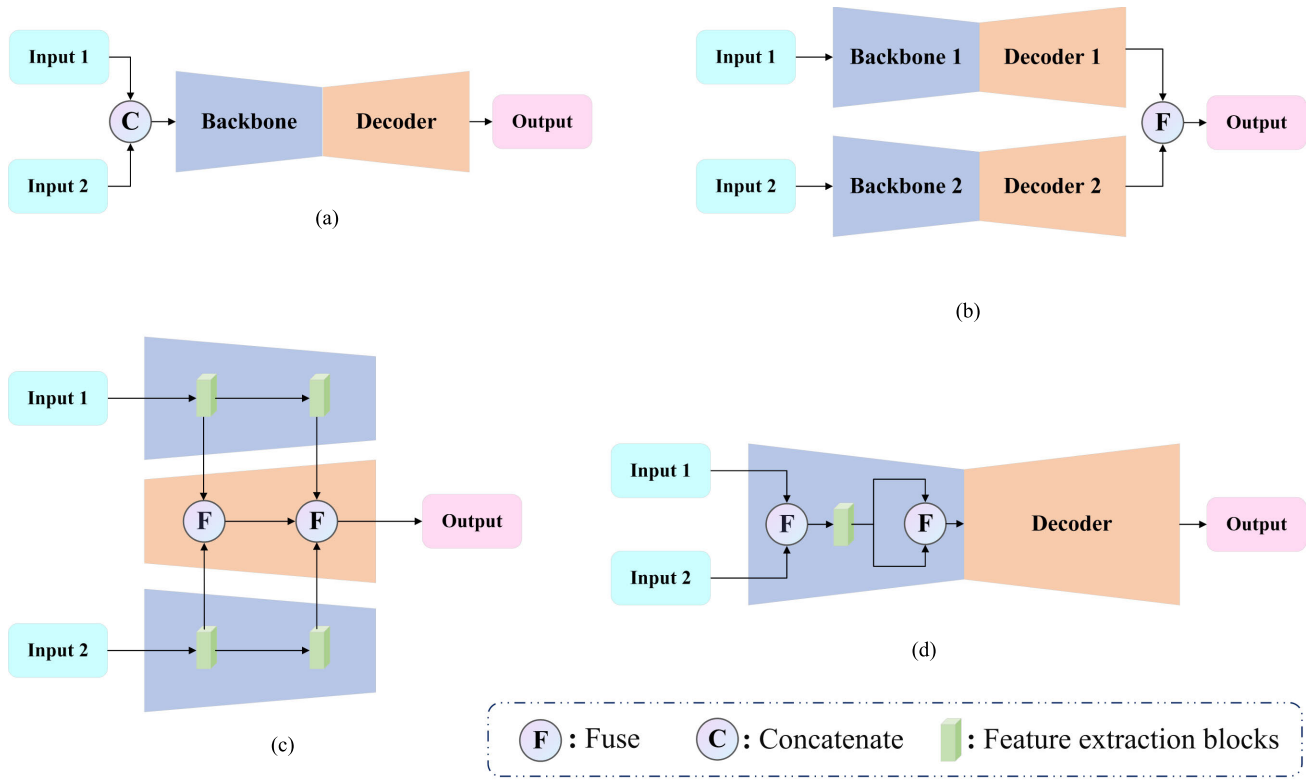


Fig. 1. Four typical multimodal fusion paradigms. (a) Early fusion schemes, (b) late fusion schemes, (c) multibranch nonlocal fusion strategy in mid-fusion schemes, and (d) multibranch Siamese fusion strategy in mid-fusion schemes.

lack of multiscale hierarchical structure make it challenging to apply to semantic segmentation tasks requiring advanced image attributes [39]. In response, SegFormer [40] refined the granularity of a feature map and introduced a multiscale hierarchical structure, and HaloNet [41] and Swin transformer [42] developed local self-attention to control the model complexity. Given that self-attention struggles to model local semantics effectively, CMT [43] and CvT [44] combine convolution with self-attention. Some works, such as MANet [45], EGENet [46], and UNetFormer [47], further validate the effectiveness of these two operators in remote sensing image semantic segmentation tasks. In addition, significant explorations have been conducted into the efficacy of variant convolutions [48], [49], [50]. By coupling these advanced components with the semantic segmentation decoder, exciting performance can be achieved in many monomodal semantic segmentation benchmarks. However, there is still considerable exploration space for multimodal semantic segmentation.

B. Multimodal Semantic Segmentation Methods

Multimodal semantic segmentation networks can provide supplementary clues for a network's ambiguous decisions through the introduction of other perceptual mode information from the data source [15], [51], [52]. The core of multimodal semantic segmentation lies in how to fuse cross-modal inputs, which are primarily categorized into early fusion [see Fig. 1(a)], late fusion [see Fig. 1(b)], and mid-fusion [see Fig. 1(c) and (d)].

Early fusion refers to concatenating multimodal data along the channel dimension before feeding them into a network [10], [12], [53]. However, this data fusion pattern simply fuses cross-modal information from the network's first linear layer, treating multimodal data equally, ignoring the inherent differences in information entropy between different modalities, and lacking nonhomogeneous expressive capabilities. For cross-modal data with significant distribution differences, the early fusion approach cannot fully leverage diverse information sources. Late fusion refers to employing a separate encoder–decoder for each multimodal data and performing fusion in the later stages of the network [54], [55]. In this fusion method, there are insufficient information interactions between cross-modal data, and the rich information from each modality gradually becomes compressed or even lost. After addressing the issues of early fusion and late fusion, researchers have largely shifted toward employing mid-fusion, which involves completing fusion before the decoder.

Mid-fusion involves two primary strategies.

1) *Multibranch Nonlocal Fusion Strategy [11], [16], [17]:* In this approach, independent feature extraction backbones are assigned to each multimodal input during the encoding phase, and cross-modal multilevel features are then fused at the decoding end. This fusion strategy accounts for the distribution differences of cross-modal data, revealing its competitive performance.
2) *Multibranch Siamese Fusion Strategy [15], [18]:* The main distinction of this fusion strategy is that the feature extraction backbone for each multimodal input shares weights. However, this approach adopts homogeneous

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operators for cross-modal features, and their inherent differences are ignored. Different types of data weights are treated equally, making similar contributions to segmentation and performing poorly when handling data with significant distribution differences.

In addition, most studies of these two mainstream mid-fusion modes have mainly considered limited dual-input cases. For the multibranch nonlocal fusion strategy, adding each extra new modality input necessitates the allocation of a new backbone since the backbone is the heaviest part of a model, leading to a sharp increase in Params, computational complexity, and inference time. Especially in the era of widespread popularity for large-scale ViTs, this linear cost leads to a burden that consumer-grade computer hardware can hardly afford. On the other hand, in the Siamese fusion strategy, the backbone part with weight sharing remains computationally intensive. Although the Params will not change, this portion still needs to be computed multiple times, requiring a similar increase in computational and time complexity. In summary, the extrapolation of existing bimodal paradigms is at the cost of high incremental computational complexity. In this study, to achieve "green AI," we aim to control the model complexity while seeking an efficient universal solution that can accomplish semantic segmentation of arbitrary quantities of multimodal data.

III. Methodology

In this section, we introduce the proposed MultiSenseSeg; as shown in Fig. 2, it consists of multiple independent MSEs, an AMM module, a feature extraction pipeline, a decoder neck, a decoding gate (DG), and an auxiliary classifier. The detailed overall structure is illustrated in Fig. 3; the multimodal backbone formed by MSEs, AMM, and the pipeline is the core part of the entire network. It can generate four multiscale feature maps of different resolutions, which are subsequently fed into the decoder part. Apart from the network's main path, the auxiliary classifier is added at the shallow layer to prevent feature degradation.

When designing our model to control extrapolation cost increments, we mainly focus on the following two metrics.

1) *Params:* An intuitive metric measuring the model complexity directly expresses the parameter scale of a model. For instance, the Params of a 1×1 convolutional kernel is 1 B. Generally, a larger number of model parameters results in a higher fitting capacity, leading to increased precision. However, this also implies the necessity for greater computational cost.
2) *Floating Point Operations Per Second:* Floating point operations per second (FLOPs) represent the number of matrix floating-point multiplications carried out by the model. As most of the computational load in a model arises from matrix multiplication, FLOPs reflect the model's computational complexity. Generally, under similar parallelism, a model with high FLOPs results in a slow inference speed and requires powerful hardware loads. The FLOPs are quadratic in relation to the image resolution.

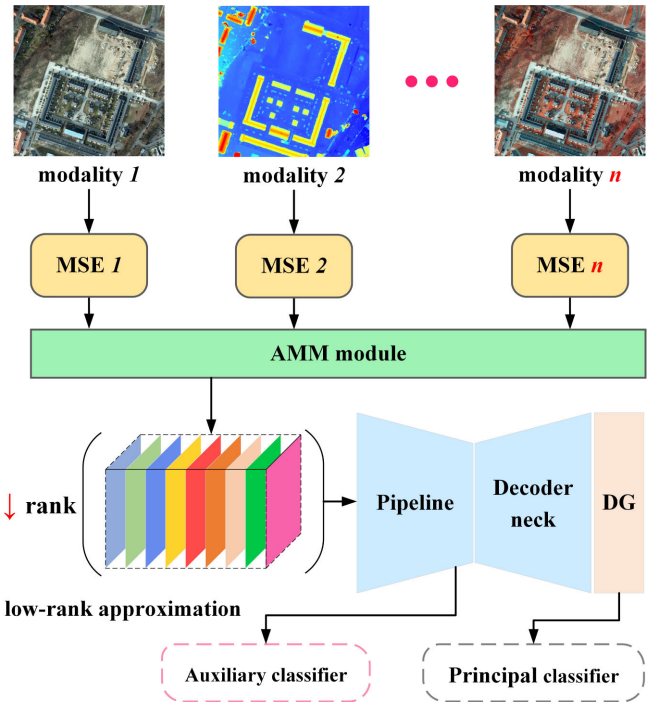


Fig. 2. Overall block diagram of MultiSenseSeg.

A. Modality-Specific Experts

Multiple MSEs share a consistent structure but have independent weights. They are employed to learn specific distributional features of intramodal while avoiding direct mixing modalities with large gaps that could deteriorate the features. Considering the linear computational load, we adopt a lightweight design for this part.

The ith MSE receives input signals from the ith modality. As shown in Fig. 4, given a parameterized matrix $X_i \in \mathbb{R}^{C_i \times H \times W}$, the ith MSE first maps it to a higher order form by using convolutions. A crucial subsequent design is the modality class token, which can provide a unique identifier for each modal category in the form of inductive bias. Specifically, we use a truncated cosine function within the interval $[0, 3/4\pi]$ to initialize the modality class tokens. Within this interval, the function possesses both injectivity and asymmetry. Injectivity can ensure that the initialization is unique and nonrepetitive, while asymmetry prevents mutual cancellation among multimodal features in some special situations. We attribute gradient properties to the modality class token and sample it through matrix broadcasting to match the scale of the feature maps. This enables the tokens to be learned by the network during the training process. Notably, our model can automatically discard the modality class token in the monomodal case.

Next, to diversify the extraction of intramodal features, we employ multigroup convolutions (MGCs). The feature map is divided into G groups along the channel dimension, and each group shares a set of convolutional kernel weights while not communicating across groups. MGCs can reduce the computational cost, with Params and FLOPs being only $1/G$ of the standard convolutions. Subsequently, a pointwise

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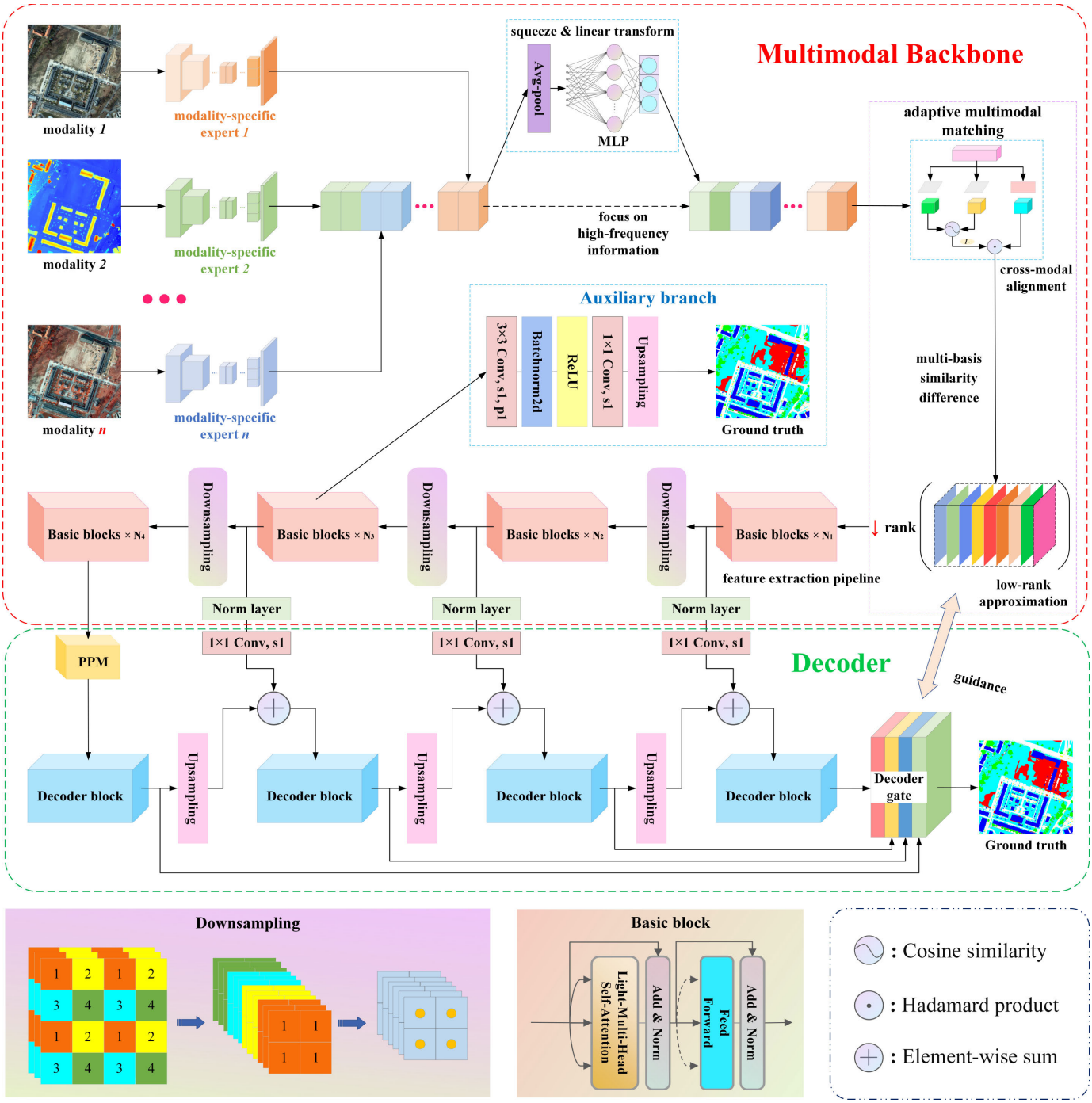


Fig. 3. Overall structure diagram of the proposed MultiSenseSeg, where $N_1 = N_2 = N_4 < N_3$ in the feature extraction pipeline.

convolution is employed to fuse the intergroup information, followed by a residual connection. This process outputs a higher order feature map $X_i^{'} \in \mathbb{R}^{C\times H\times W}$. The Params and FLOPs required for this part of n different multimodal inputs are given as follows:

$$\text{Params} = \left(\frac{9}{4G} + 1\right) nC^2 + n + \sum_{i=1}^{n} 9C_i C \tag{1}$$

$$\text{FLOPs} = \left(\frac{9}{4G} + 1\right) nHWC^2 + \sum_{i=1}^{n} 9HWC_i C \tag{2}$$

where G denotes the number of groups, C_i denotes the number of channels for the ith modality, H and W denote the height and width of the data, respectively, and C denotes the output channel number ($C > C_i$).

Due to the inevitable inclusion of unnecessary information in feature maps (usually represented as lower-frequency channels with more negative values), not suppressing this information during extrapolation will lead to the production of suboptimal features. Hence, we utilize channel attention to focus on high-frequency information, selecting high-quality intramodal feature bases. The calculation process of the

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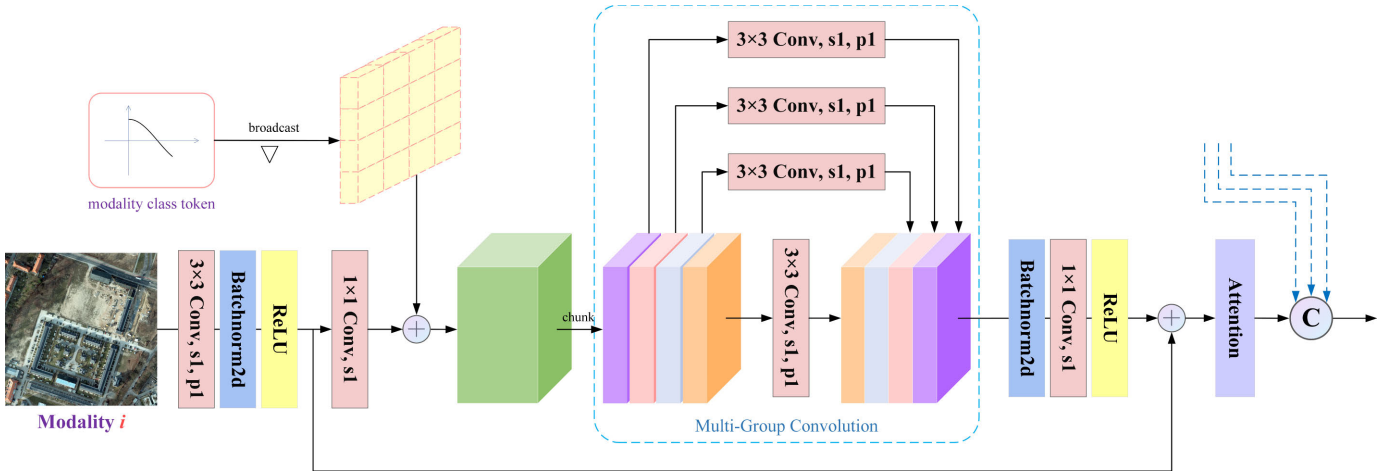


Fig. 4. Structural diagram of the MSEs.

Algorithm 1 Pseudocode of the MSEs' Workflow

Input: Multimodal data: $Modality_1 = \{x_{11}, x_{12}, \ldots, x_{1|x|}\}$, $Modality_2 = \{x_{21}, x_{22}, \ldots, x_{2|x|}\}$, $Modality_3 = \{x_{31}, x_{32}, \ldots, x_{3|x|}\}$;
hyperparameter: $In_chans = [3, 1, 4]$ // change this to perform modal extrapolation

```
MSEs = []
for i ← 0 to in_chans.length: do
    append MSE(input_channels = in_chans[i], ...) to MSEs
end for
for x1, x2, x3 in Modality_1, Modality_2, Modality_3: do
    All_modalities, Out_feature_maps = [x1, x2, x3], []
    for j ← 0 to MSEs.length: do
        out_feature_map = MSEs[j](All_modalities[j])
        append out_feature_map to Out_feature_maps
    end for
    concatenate Out_feature_maps along 2nd dimension
    then feed into AMM module
end for
end
```

attention map $U \in \mathbb{R}^{C\times 1\times 1}$ is given as

$$U = \sigma\left(\mathrm{lin}^{(2)}\left(\mathrm{act}\left(\mathrm{lin}^{(1)}\left(\frac{\sum_{i=1}^{H}\sum_{j=1}^{W} X(i,j)}{HW}\right)\right)\right)\right) \quad (3)$$

where $\sigma(*)$, $\mathrm{lin}(*)$, and $\mathrm{act}(*)$ denote the sigmoid function, linear transformation, and activation function, respectively. The scores of U are assigned to the corresponding elements of X, thereby compelling the network to focus on the highly contributing information.

Notably, to avoid the manual burden of altering the model structure during extrapolation, we control this functionality through only a single model hyperparameter. For simplicity, we take situations of simultaneously inputting three modalities with channels $= \{3, 1, 4\}$ as an example. The pseudocode of the MSEs workflow is illustrated briefly in Algorithm 1. Given the set of channel numbers for each modality, MultiSenseSeg automatically identifies the number of modalities to be processed and initializes specific MSEs based on the dimensions of each modality. The outputs of different MSEs are then concatenated along the channel dimension and fed into the AMM module. At this point, we obtained cross-modal higher order features and buffered the modality gaps.

B. Adaptive Multimodal Matching

Different from monomodal and bimodal cases, the following challenges arise when dealing with quantity-agnostic modalities: 1) how to search for complementary features in n modes and 2) how to carry modal semantics in a low-dimensional form. Specifically, repetitive stacking of similar feature bases cannot provide new perceptual information for the network; however, an effective approach to maximally drive heterogeneity among modalities from global modality categories is lacking. In addition, multimodal feature maps have high-order data structures, which reduce network inference efficiency. Furthermore, because of the field-of-view disparities in photogrammetric sensors, the same-named pixels across different modalities are not always strictly aligned, which will lead to phase differences in subsequent spatial

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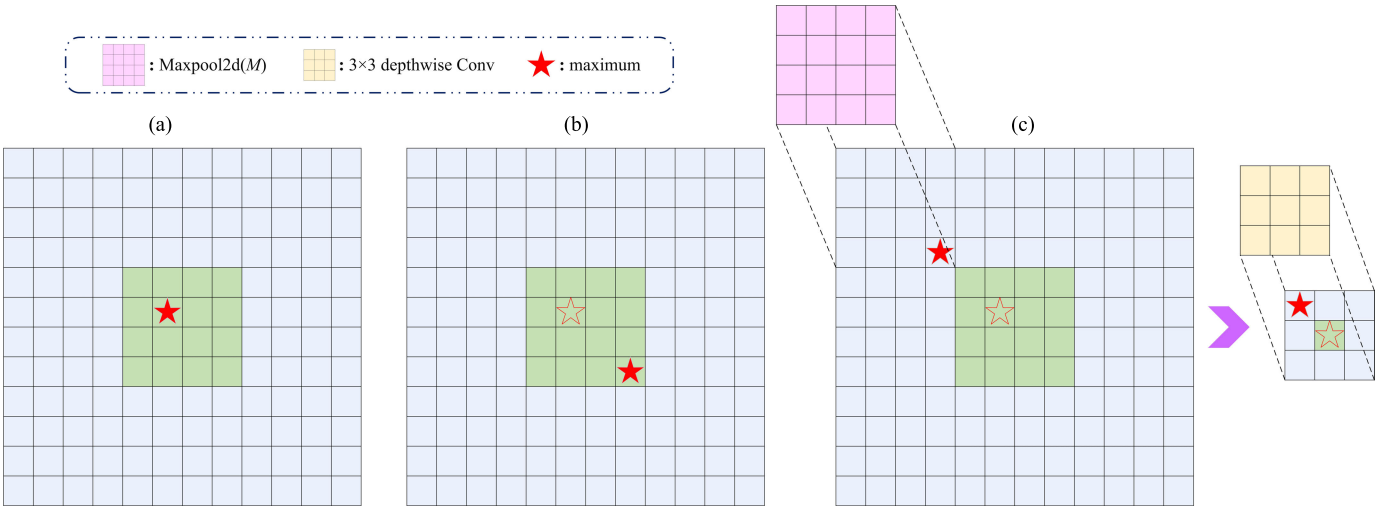


Fig. 5. Illustration of coordinate offset correction. (a) Reference image without any field-of-view offset. (b) and (c) Two different instances of offset.

steps and reduce operation reliability. To address these issues, we design an AMM module, which can be refined into two subprocesses: cross-modal alignment and multibasis similarity difference.

For clarity, we use $X = [X_1; X_2; \ldots; X_n]$ to denote the high-order multimodal matrix received by AMM, where the stacks of n modalities are arranged solely along the channel dimension without fusion. Each modality can be decomposed into m individual subspaces in the channel dimension, denoted as $X_i = [X_{i1}^{\mathrm{T}}, X_{i2}{}^{\mathrm{T}}, \ldots, X_{im}^{\mathrm{T}}]$, with each subspace representing a basis for the matrix. The entire multimodal matrix $X \in \mathbb{R}^{\mathcal{C}\times H\times W}$, $\mathcal{C} = n \times m$, can be redefined as follows:

$$X = \left[X_{11}, \ldots, X_{1m}; \ldots, X_{ij}, \ldots; X_{n1}, \ldots, X_{nm}\right]^{\mathrm{T}} \tag{4}$$

where the basis $X_{ij}^{\mathrm{T}} \in \mathbb{R}^{1\times(H\times W)}$, $1 \le i \le n; 1 \le j \le m$, denotes a row vector, representing the 1-D sequence form of the jth channel feature map of the ith modality.

For the cross-modal alignment, we first replicated X into three unequal copies through linear transformations, with a set of weights for each modality, expressed as

$$\begin{aligned}\left\{\tilde{X}_1, \tilde{X}_2, \tilde{X}_3\right\} &= X\left[\mathcal{W}_1; \mathcal{W}_2; \ldots; \mathcal{W}_n\right], \quad \text{where} \\ \mathrm{len}(\mathcal{W}_i) &= 3m.\end{aligned} \tag{5}$$

Subsequently, these replicated copies are split into multiple $M \times M$-sized patches and then compute a representative value within each patch as a local descriptor. Thus, even if there is an overall shift in an image, as long as the scope of establishing injective relations with the representative value's position remains within the $M \times M$ range, the alignment of multimodal key information can be roughly maintained. A representative value within a local range is often regarded as the maximum value or a weighted sum with respect to the neighborhood. Hence, for $\tilde{X}_1$ and $\tilde{X}_2$, we employ max-pooling and a 3×3 depthwise convolution to extract the key value $\mathcal{P}_k$ from the patch at position k. This process can be expressed as

$$\mathcal{P}_k = \sum_{h=-1}^{1}\sum_{w=-1}^{1} w_{\mathrm{offset}(h,w)\to k} \cdot \mathrm{Max}\left(\mathcal{N}_8\left[\mathcal{P}_k\right]\right) \tag{6}$$

where $\mathcal{N}_8[\mathcal{P}_k]$ denotes the eight-neighborhood range centered around the patch at position k, $\mathrm{Max}(\cdot)$ denotes the max-pooling operation, and $w_{\mathrm{offset}(h,w)\to k}$ denotes the weight in the convolutional kernel at the relative offset of (h, w) to the central position k. Through successive operations of max-pooling and weighted convolution, the network blurs the values within a patch into $\mathcal{P}_k$. As illustrated in Fig. 5, for the case where the maximum value remains within the patch [see Fig. 5(b)], consistency with the unshifted image [see Fig. 5(a)] is maintained after max-pooling, while for the case where the maximum value is offset out of the patch [see Fig. 5(c)], the image can also be aligned by combining it with the neighborhood of convolution. For $\tilde{X}_3$, to better preserve fine-grained semantics, we employ a large convolution kernel instead. Since both max-pooling and convolution operations are learnable, the value $\mathcal{P}_k$ can be dynamically adjusted through gradient backpropagation.

The workflow of the proposed multibasis similarity difference has clear physical significance, i.e., the cosine distance in space. The greater the cosine distance between two bases is, the more likely they are complementary features. We first calculate the cosine similarity matrix $\mathcal{S}$ as follows:

$$\mathcal{S} = \mathrm{Sim}\left(\tilde{X}_1, \tilde{X}_2\right) = \lambda \frac{\tilde{X}_1 \odot \tilde{X}_2}{\left\|\tilde{X}_1\right\|_2 \left\|\tilde{X}_2\right\|_2} + \tilde{\mathcal{B}} \tag{7}$$

$$\tilde{\mathcal{B}} = \mathrm{lin}^{(2)}\left(\mathrm{ReLU}\left(\mathrm{lin}^{(1)}\left(\mathrm{sign}(\mathcal{B}) \cdot \log_2(1 + |\mathcal{B}|)\right)\right)\right) \tag{8}$$

$$\mathrm{sign}(\mathcal{B}) = \begin{cases} 1, & \mathcal{B} > 0 \\ 0, & \mathcal{B} = 0 \\ -1, & \mathcal{B} < 0 \end{cases} \tag{9}$$

where $\mathcal{S} \in \mathbb{R}^{\mathcal{C}\times\mathcal{C}}$; the value at the lth row and rth column indicates the similarity score between the lth basis when it is the focus and the rth basis $(l, r < \mathcal{C})$; $\odot$ denotes the Hadamard product; λ is a learnable scaling factor used to dynamically adjust the contribution ratio initialized to 10; and $\tilde{\mathcal{B}}$ denotes the relative positional biases between bases and is designed to simulate the arrangement order of the bases. Specifically, we define that when the focus is on the

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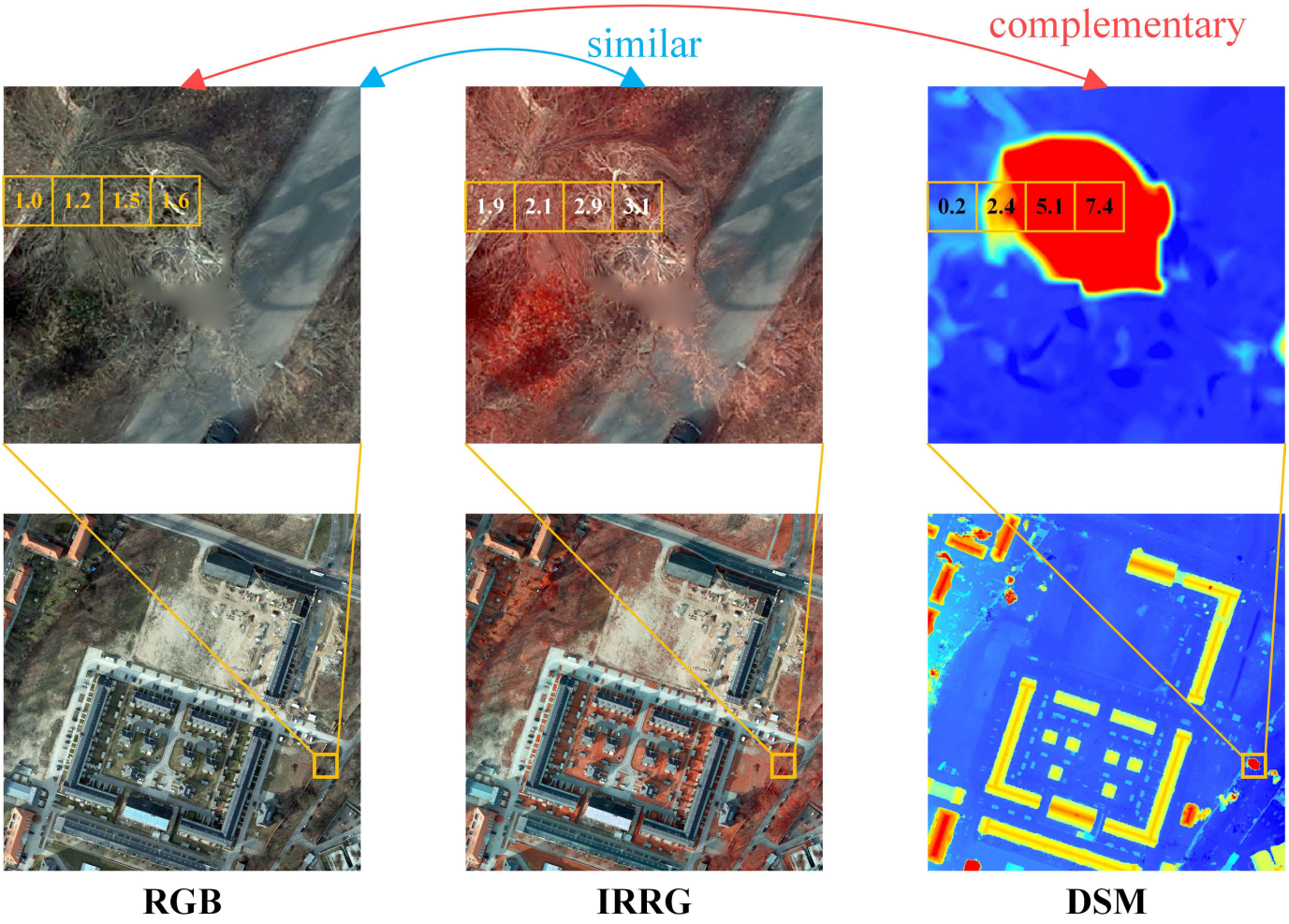


Fig. 6. Example of the multibasis similarity difference mechanism, where the constant values represent the pseudo-eigenvalues of the pixel at the corresponding position of the jth basis in that modality.

lth basis, the relative position of the rth basis is $r - l$. The AMM initializes a matrix $\mathcal{B}$ containing relative position coordinates following the aforementioned rule, projects it to logarithmic space, and then applies a series of transformations to obtain $\tilde{\mathcal{B}}$. Notably, projecting it to the logarithmic space is to reduce the extrapolation ratio. For an example of a feature map processed by a single MSE with 32 channels, when the input increases to three modalities, the relative position range will be from $[-31, 31]$ to $[-95, 95]$, with an extrapolation ratio of $((95 - 31)/31) = 2.065$, while, in the logarithmic space, the dynamic range changes will be from $[-5, 5]$ to $[-6.585, 6.585]$, with an extrapolation ratio of only 0.317, which is much smoother than that in linear space. This approach decouples the steepness of $\mathcal{B}$ from the total number of channels, which is especially important when more modalities are needed. Next, we calculate the cosine distance and assign it to the corresponding base

$$\hat{X} = \text{Softmax}^{(2)}\left(1 - \text{Softmax}^{(1)}(\mathcal{S})\right) \odot \tilde{X}_3 \tag{10}$$

where $1 - \text{Softmax}^{(1)}(\mathcal{S})$ is the cosine distance matrix, which transforms similarity into differences. After another $\text{Softmax}^{(2)}(*)$ operation, the normalized cosine distance is obtained, and then, it is dot multiplied by $\tilde{X}_3$ to assign the heterogeneity relation correspondences to the bases. By these, bases with low correlation are considered complementary information by the network and are assigned higher scores. Subsequently, we employ global convolutions along the channel path to mix the basis features, perform matrix rank reduction, and filter redundant information. To prevent feature degradation, we also introduce residual connections. The entire operation can be approximated as

$$\mathcal{G}(X) = \text{Softmax}^{(2)}\left(1 - \text{Softmax}^{(1)}\left(\lambda \frac{\tilde{X}_1 \odot \tilde{X}_2}{\|\tilde{X}_1\|_2 \|\tilde{X}_2\|_2} + \tilde{\mathcal{B}}\right)\right) \odot \tilde{X}_3 + X, \quad \text{where } \{\tilde{X}_1, \tilde{X}_2, \tilde{X}_3\} = X\mathcal{W}. \tag{11}$$

This is a dynamic adaptive aggregation process in which the network can learn exclusive implicit connection weights W between modalities from concrete instances. For better understanding, let us illustrate this inference process with a simple example

As shown in Fig. 6, there are three types of multimodal remote sensing images, RGB, IRRG, and DSM, and the corresponding marker positions include two classes: "low

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vegetation" and "tree." These two classes have similar visual representations in optical images but significant differences in height. We assume that the values of the marked positions of the jth basis in the three modalities can be represented by the constants in the figure. Since the recognition of the two classes in RGB and IRRG images is similar, the vector representations of their jth bases may differ mainly in vector length while maintaining almost the same direction. In contrast, the vector direction for DSM images can be considered entirely different since stacking homogeneous bases repeatedly cannot provide the network with distinct perceptual dimensions of information. For this case, the proposed multibasis similarity difference mainly performs operations outlined in (12), as shown at the bottom of the page, i.e., when focusing on x_{1j}^{T}, more attention is directed toward x_{3j}^{T}, while x_{2j}^{T} is discarded. In this way, the original feature matrix can be well approximated by a low-rank matrix [rank(3) $\rightarrow$ rank(2)]. This is a best-of-both approach, which not only locates complementary information but also reduces the computational complexity of the subsequent feature extraction step.

C. Feature Extraction Pipeline

1) Architecture Deployment: To interpret the multimodal semantic features from multiple scales, we divide the feature extraction pipeline into four different resolution levels. Adjacent levels take 2 as a scale factor, broadening the network width while downsampling. The downsampling strategy entails interleaved sampling in the row and column dimensions of the feature maps. Subsequently, concatenation is applied along the channel dimension, and compression is applied through a linear projection to reduce the number of channels ($\dot{X} \in \mathbb{R}^{\mathcal{C}\times H\times W} \rightarrow \ddot{X} \in \mathbb{R}^{4\mathcal{C}\times (H/2)\times (W/2)} \rightarrow \ddot{X} \in \mathbb{R}^{2\mathcal{C}\times (H/2)\times (W/2)}$). Each level is equipped with a certain number of identical basic blocks. We stack the same number of blocks in the first, second, and fourth levels while stacking more blocks in the third level (denoted as $\{N_1, N_2, N_3, N_4\}$, $N_1 = N_2 = N_4 < N_3$).

2) Operator Design: Long-range dependencies and local contextual relationships are crucial for feature extraction. In SOTA monomodal semantic segmentation approaches, two operators, self-attention and convolution, are usually employed to address these two perceptual demands, respectively. However, multimodal feature maps contain richer information dimensions, which implies that multimodal extrapolation may rely on wider network architectures to accommodate varied feature bases for comprehensive feature extraction. To better adapt to multimodal cases, we design two variant forms of operators, light-window multihead self-attention (LW-MHSA) and multigroup depthwise separable convolution (MG-DSC), to reduce the computational cost of modal extrapolation. The LW-MHSA is a dynamic spatial aggregation operator employed to model long-distance dependencies, while the MG-DSC is a static operator with inductive bias that is able to extract local information.

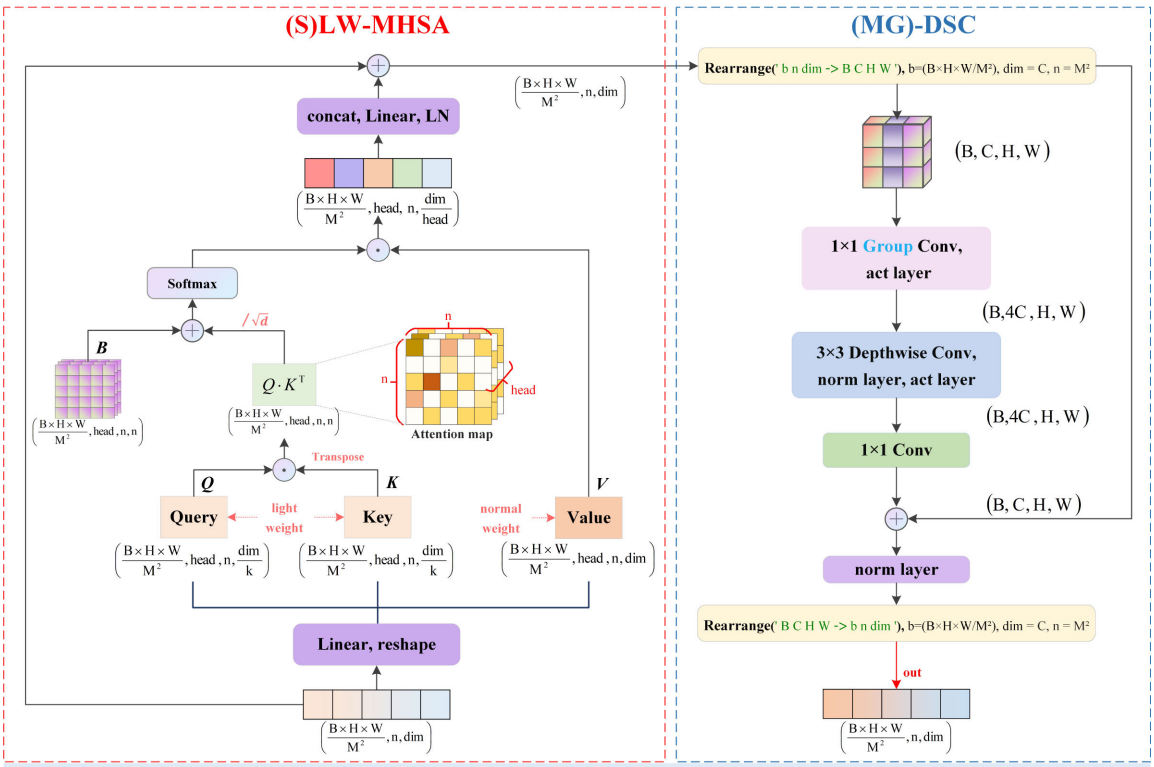
For the LW-MHSA, self-attention is deployed based on local windows. We first partition the feature map $X \in \mathbb{R}^{\mathcal{C}\times H\times W}$ into (HW/M^2) nonoverlapping small patches, each with a length and a width of M. Next, the feature map is projected into three implicit state matrices, query (Q), key (K), and value (V), using different linear weights. Specifically, Q and K are sparsified to $1/k$ of their previous dimensions to make them more lightweight, while V maintains the same order. After stretching Q, K, and V into 1-D sequences along the spatial path, self-attention is performed within each window. Each vector is further divided into h heads along the channel path, where each head represents a single-dimensional subspace of the original multimodal map, focusing on a specific semantic dimension. The process of attention in the ith head can be formulated as (13), shown at the bottom of the page. Here, $\mathcal{W}_{*[i]}^{(L)}$ and $\mathcal{W}_{*[i]}^{(N)}$ denote the light weights and normal weights, respectively. The sparsification of Q and K is performed directly within the linear transformation without relying on additional downsampling operations, which can accelerate the network painlessly. For the MG-DSC, we first restore the 1-D sequence back to $X' \in \mathbb{R}^{\mathcal{C}\times H\times W}$ and then feed it into multigroup pointwise convolution to expand the channel dimension. Subsequently, we perform independent depthwise separable convolutions within each group to achieve diverse feature translation across multiple groups for the multimodal feature maps. Ultimately, these features are mixed in the channel dimension in the form of weighted sum decisions. The entire process can be referenced in Fig. 7(a).

Although the aforementioned designs are efficient, they may introduce information flow patterns that are not conducive to feature learning. Specifically, in the LW-MHSA, attention is performed within each window; since the weights are only shared along the channel dimension, information exchange

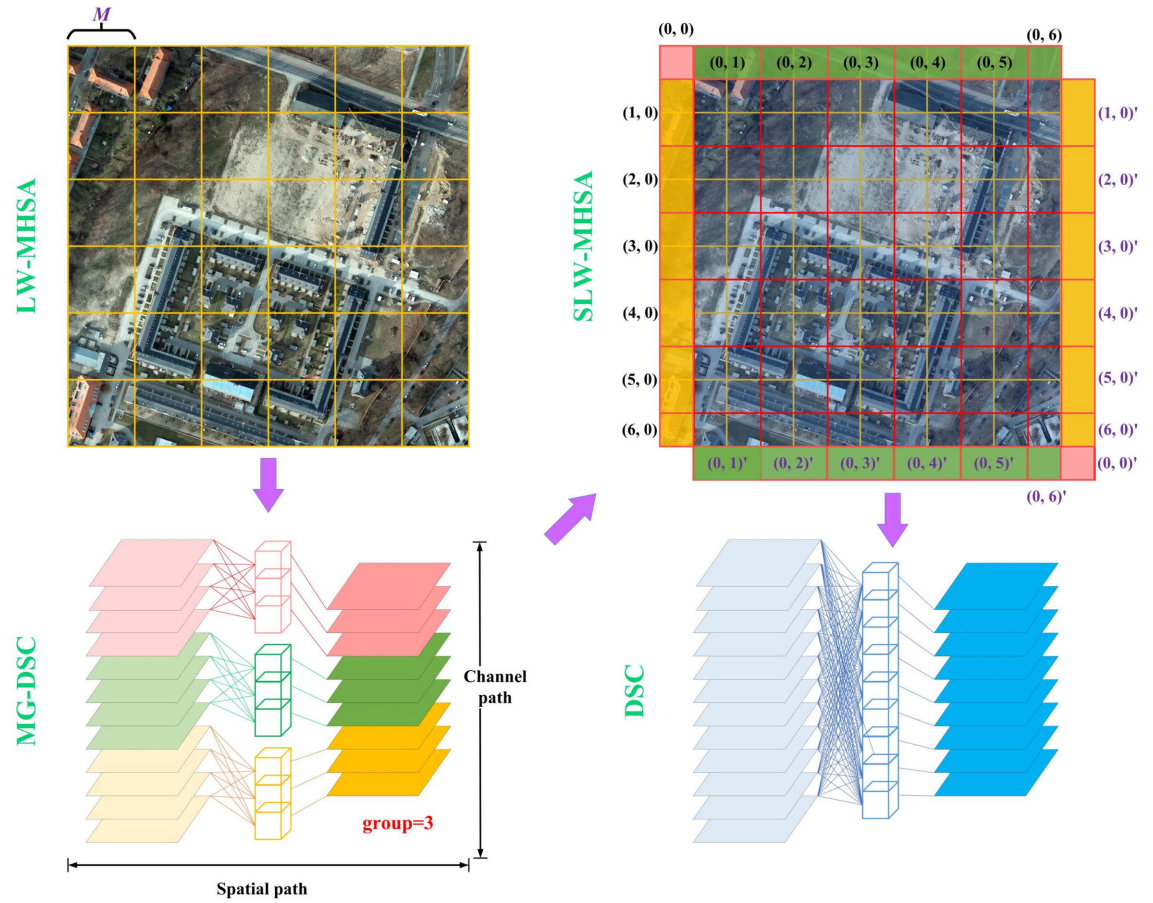
$$\begin{bmatrix} x_{1j}{}^{\mathrm{T}} \\ x_{2j}{}^{\mathrm{T}} \\ x_{3j}{}^{\mathrm{T}} \end{bmatrix} = \begin{bmatrix} \mathrm{RGB}_j \\ \mathrm{IRRG}_j \\ \mathrm{DSM}_j \end{bmatrix} = \begin{bmatrix} 1.0 & 1.2 & 1.5 & 1.6 \\ 1.9 & 2.1 & 2.9 & 3.1 \\ 0.2 & 2.4 & 5.1 & 7.4 \end{bmatrix} \sim \begin{bmatrix} 1.0 & 1.0 & 1.5 & 1.5 \\ 2.0 & 2.0 & 3.0 & 3.0 \\ 0.0 & 2.5 & 5.0 & 7.5 \end{bmatrix} \propto \begin{bmatrix} 2 & 2 & 3 & 3 \\ 2 & 2 & 3 & 3 \\ 0 & 1 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 3 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \end{bmatrix} \tag{12}$$

$$\mathrm{LW\text{-}MHSA}\left(X_{[i]}\right) = \mathrm{Softmax}\left(\frac{\left(X_{[i]}\mathcal{W}_{Q[i]}^{(L)}\right) \odot \left(X_{[i]}\mathcal{W}_{K[i]}^{(L)}\right)^{\mathrm{T}}}{\sqrt{\mathcal{C}/h}} + B\right) \odot \left(X_{[i]}\mathcal{W}_{V[i]}^{(N)}\right) \tag{13}$$

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(a) Structure of the basic block



(b) Calculation process in the basic block of the odd (left) and even (right) indices.

Fig. 7. Illustration of the basic block. (a) In MG-DSC, group convolution is exclusively applied to blocks with odd indices, whereas regular standard convolution is employed for blocks with even indices (M denotes the window size and C denotes the number of channels in the 2-D feature map). (b) (i, j)' of the SLW-MHSA represents the position (i, j) after shifting.

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across windows in the spatial path is hindered. In the MG-DSC, the combination of MGC and depthwise convolutions, both of which are based on sparse sampling, could also restrict information exchange in the channel path. To address this, as shown in Fig. 7(b) in the LW-MHSA, we introduce a cyclic shift window mechanism, i.e., in the subsequent basic block, the window positions are shifted to the lower right by $M/2$ to enhance communication with neighboring windows. Meanwhile, for the MG-DSC, replacing the first convolutional layer with a standard convolution (group = 1) in even-indexed blocks can effectively alleviate the issue of inadequate information communication in the channel path. The process can be expressed as follows:

$$\begin{aligned}\hat{X}_l &= \text{LW-MHSA}(\text{LN}(X_{l-1})) + X_{l-1}\\ X_l &= \text{MG-DSC}\big(\text{Norm}\big(\hat{X}_l\big)|_{\mathcal{G}}\big) + X_l\\ \hat{X}_{l+1} &= \text{SLW-MHSA}(\text{LN}(X_l)) + X_l\\ X_{l+1} &= \text{DSC}\big(\text{Norm}\big(\hat{X}_{l+1}\big)\big) + X_{l+1}\end{aligned} \tag{14}$$

where $\hat{X}_l$ and X_l denote the outputs of the (S)LW-MHSA and the (MG)-DSC for block l, respectively. SLW-MHSA denotes the window attention after cyclic shifting, $\text{MG}-\text{DSC}(*|_{\mathcal{G}})$ denotes the MG-DSC with a group number of $\mathcal{G}$, and $\text{DSC}(*)$ denotes the MG-DSC with only a single group. Through the aforementioned combination, it is ensured that the feature information of multiple modalities interacts fully in both the spatial and channel dimensions.

The FLOPs of the two existing mainstream self-attention mechanisms, traditional multihead self-attention (MHSA) and window-based MHSA (W-MHSA), as well as our LW-MHSA, can be approximated as follows:

$$\text{FLOPs}_{(\text{MHSA})} = 4HW\mathcal{C}^2 + 2(HW)^2\mathcal{C} \tag{15}$$

$$\text{FLOPs}_{(\text{W-MHSA})} = 4HW\mathcal{C}^2 + 2M^2HW\mathcal{C} \tag{16}$$

$$\text{FLOPs}_{(\text{LW-MHSA})} = \left(2+\frac{2}{k}\right)HW\mathcal{C}^2 + \left(1+\frac{1}{k}\right)M^2HW\mathcal{C} \tag{17}$$

where $M^2 \ll HW$. It is obvious that for any $k > 1$, the proposed LW-MHSA requires the least computational complexity. Notably, since memory-bound Q-K length, their long-sequence memory complexity (which refers to the GPU memory required for storing long multimodal sequences) also corresponds to $O(N^2)$, $O(N)$, and $O(N^c)$ $(0 < c < 1)$, respectively. For the MG-DSC, its FLOPs are only $1/g$ of the standard depthwise separable convolution. In addition, from the perspective of algorithmic mechanisms, these sparsification designs can be regarded as regularization approaches, reducing the connectivity density of neurons in the network's hidden layers and lowering the model complexity without decreasing the hidden representation size. This implies that the risk of overfitting is reduced without reducing the quantity of rich multimodal feature representations in the network, allowing for the learning of more robust multimodal visual representations.

D. Decoder

Referring to SOTA methods in monomodal semantic segmentation, we employ UperNet [56] as the decoder and enhance it with a DG to optimize information flow. UperNet is composed of a pyramid pooling module (PPM) [29] and a feature pyramid network (FPN) [23]. The DG is a second-order operation that enhances the deep spatial indexing capabilities. Specifically, considering the abstract semantics carried by deep multimodal feature maps, we use shallow feature maps that retain rich spatial details to guide segmentation, which can be formulated as follows:

$$\mathcal{X} = X_{\text{de}} \cdot \sigma\big(\text{ReLU}(X_{\text{de}}\mathcal{W}_{\text{de}} + X_{\text{en}}\mathcal{W}_{\text{en}})\cdot W_{\varphi}\big) \tag{18}$$

where X_{de} and X_{en} denote the feature maps from the FPN cascade at the decoding end and the output of the AMM at the encoding end, respectively. $\text{ReLU}(*)\cdot\mathcal{W}_{\varphi}$ aims to enhance the network's nonhomogeneous expressive capacity and generate an attention score $\varphi \in \mathbb{R}^{1\times H\times W}$. We parallelize this part with channel attention, suppressing irrelevant feature responses on a global scale. The decoding feature maps are then connected to the classifier, and the final segmentation map is the output, $X_{\text{out}} \in \mathbb{R}^{n\text{classes}\times H\times W}$.

E. Loss Function

To prevent the degradation of multimodal information during long gradient flow, an auxiliary classifier is added to the middle layer. Considering that the third level of the feature extraction pipeline is the primary stage for feature extraction, we add it at this location. During the initial stages of training from scratch, discrepancies between the confidence distribution of the model outputs and the ground truth are too large, requiring rapid convergence toward an appropriate gradient descent direction. Hence, shortcut learning is necessary. However, as training progresses into later stages, the excessive emphasis on shallow auxiliary branches can lead to the neural network becoming lazy and overly reliant on shortcuts. For the above reasons, we let the loss function dynamically adjust the proportion allocation based on the total number of iterations during training, i.e., allocating a larger proportion of loss to the shallow auxiliary branches in the early stages of training and gradually decreasing this proportion as training advances. The total loss can be expressed as follows:

$$\mathcal{L}_{\text{tot}} = \mathcal{L}_{\text{pri}} + \left(\frac{\beta-\alpha}{\mathcal{E}}\text{epoch} + \alpha\right)\mathcal{L}_{\text{aux}} \tag{19}$$

where $\mathcal{E}$ denotes the total number of training iterations; α and β $(\alpha > \beta)$ are two hyperparameters that denote the proportions allocated to the auxiliary branch at the beginning and end of the training, respectively; epoch denotes the current iteration round; as training progresses, the coefficient factor of the auxiliary branch $(((\beta-\alpha)/\mathcal{E})\text{epoch} + \alpha)$ gradually decreases; $\mathcal{L}$ denotes the weighted cross-entropy loss; and $\mathcal{L}_{\text{pri}}$ and $\mathcal{L}_{\text{aux}}$ denote the losses of the primary and auxiliary branches, respectively. The equation for calculating $\mathcal{L}$ when the number of classes is N can be expressed as

$$L = -\frac{1}{HW}\sum_{n=1}^{N} w_n \cdot Y_n \cdot \log_2\big(\text{softmax}\big(\hat{Y}_n\big)\big) \tag{20}$$

where Y_n and $\hat{Y}_n$ denote the ground truth and predicted values for the nth class, respectively, and w_n denotes the weight

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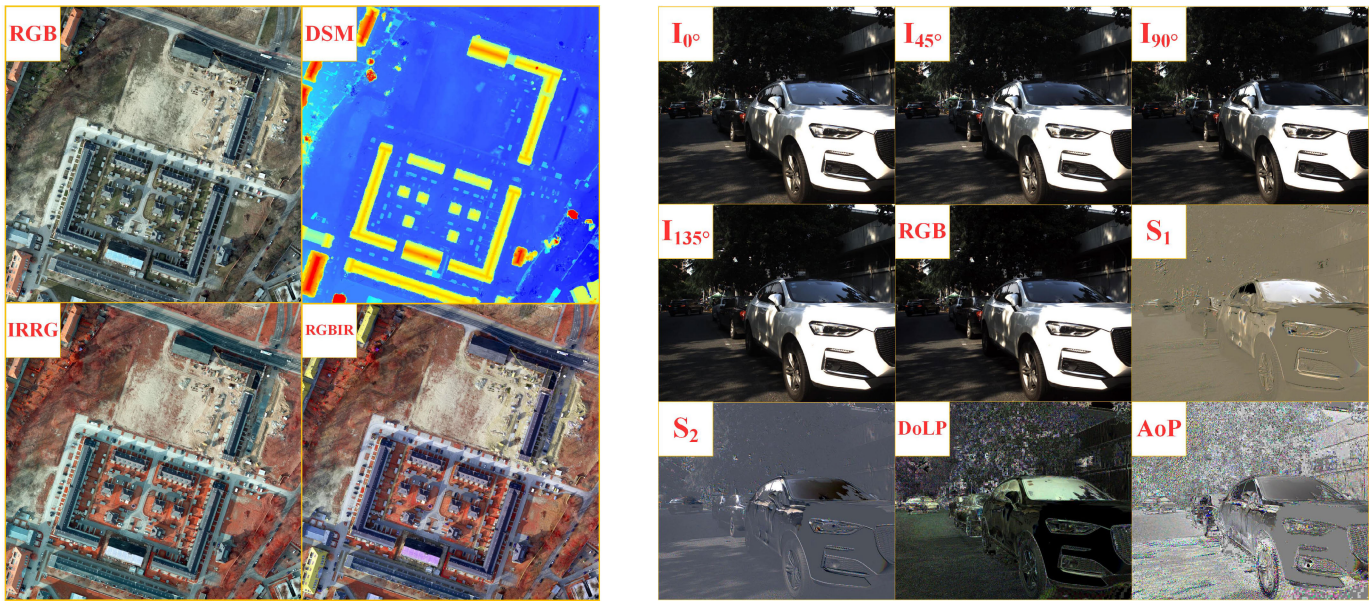


Fig. 8. Visualization of the multimodal datasets used (the ISPRS Potsdam is shown on the left, and the ZJU-RGB-P is shown on the right).

assigned to the nth class. The usage of distinct weights is aimed at mitigating class imbalance issues in remote sensing imagery.

IV. Experimental Settings

A. Datasets and Preprocessing

The ISPRS Potsdam dataset contains data from four modalities: RGB, DSM, IRGB, and RGBIR. (Indeed, the RGBIR contains information that overlaps with the previous three types. Here, we temporarily assume that its different channel combinations can yield new visual representations for subsequent experiments.) The dataset includes five foreground classes: impervious surface, building, low vegetation, tree, and car. The dataset is captured from high-resolution satellite sensors and is challenging due to the presence of small target objects, narrow streets, and dense traffic. There are a total of 38 large-scale images with a ground sampling distance of 5 cm. We eliminate the images with ID 7_10, which have annotation errors [45], [47], and the remaining images are randomly divided into training sets and validation sets at a ratio of 25:12. Each training set image is cropped into small pieces of size 512 × 512 with an overlap of 0.125; to reflect the real situation, the validation set does not overlap.

The ZJU-RGB-P dataset [52] includes eight foreground classes: building, glass, car, road, vegetation, sky, pedestrian, and bicycle. This dataset is a multimodal dataset of natural scenes. Although it is not strictly a remote sensing dataset, it can test the applicability of our model to data captured by close-range photogrammetric sensors. With the advancement of integrated "space-air-ground" remote sensing, and considering the collaborative requirements for ground-based sensors, covering such sensors is essential as a unified method. Different from small-scale objects in a bird's-eye view, the segmentation challenges of the natural scene dataset arise from the complex interplay of light and shadows, as well as specular reflections from high reflective objects. More importantly, this dataset can obtain as many as nine multimodal data with different meanings after preprocessing.

Specifically, the dataset contains data from four polarization angles (three bands): I_{0°, I_{45°, I_{90°, and I_{135°. Based on the data of the four polarization angles and the Stokes vectors, S_0, S_1, and S_2 can be obtained. Among them, S_0 describes light without any polarization, i.e., RGB imagery. The equations for calculating the Stokes vectors are given as follows:

$$\vec{S} = \begin{bmatrix} S_0 \\ S_1 \\ S_2 \end{bmatrix} = \begin{bmatrix} (I_{0^\circ} + I_{45^\circ} + I_{90^\circ} + I_{135^\circ})/2 \\ I_{0^\circ} - I_{90^\circ} \\ I_{45^\circ} - I_{135^\circ} \end{bmatrix}. \quad (21)$$

Based on the following equations, the degree of linear polarization (DoLP) and the angle of polarization (AoP) can be further obtained:

$$\text{DoLP} = \frac{\sqrt{S_1^2 + S_2^2}}{S_0} \quad (22)$$

$$\text{AoP} = \frac{\tan^{-1}\left(\frac{S_2}{S_1}\right)}{2} \quad (23)$$

where the value range of the DoLP is [0, 1]. Therefore, we set the value of the pixel beyond this range to 1. At this point, we obtain all nine multimodal data types: I_{0°, I_{45°, I_{90°, I_{135°, S_0(RGB), S_1, S_2, DoLP, and AoP. The dataset has already been divided into training and validation sets, eliminating the need for manual partitioning. Each training set image is cropped into small pieces of size 512 × 512 with an overlap of 0.25; the validation set also does not overlap. A visualization of the multimodal data in each dataset is shown in Fig. 8.

B. Models for Comparison

We evaluate the performance of MultiSenseSeg against a selection of 15 leading models from the past three years. The chosen models include both monomodal models and bimodal

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TABLE I
DETAILS OF THE MODELS FOR COMPARISON WHEN THE INPUT SIZE IS $3 \times 512 \times 512$, WHERE THE FIRST NINE ARE MONOMODAL MODELS AND THE LAST SIX ARE BIMODAL MODELS

Model	Backbone	Decoder	Params	GFLOPs
TransUNet [57]	ResNet-50 + ViT-B/16	UNet	63.386M	85.768
MANet [45]	ResNet-101	UNet	57.047M	672.230
UNetFormer [47]	ResNet-50	UNetFormer	57.230M	173.887
CMT [43]	CMT-S	UperNet	58.855M	227.633
Swin Transformerv2 [58]	Swinv2-T	UperNet	59.884M	228.626
ConvNeXtv2 [59]	ConvNeXtv2-T	UperNet	60.177M	234.476
RMT [60]	RMT-S	UperNet	58.070M	253.510
SwinUNet [61]	Swin-T$^+$	SwinUNet	62.719M	67.422
EGENet [46]	EGENet/4	EGENet	59.796M	152.441
EAFNet [52]	ResNet-34	UperNet	60.150M	147.936
NLFNet [11]	ResNet-34	UperNet	60.076M	132.367
UisNet [18]	SegFormer-B4	UisNet	64.648M	183.422
DIR_DeepLabv3+ [16]	ResNet-50	DeepLabv3+	60.461M	164.489
D_UNet [17]	UNet	UNet	60.367M	799.668
CMX [62]	SegFormer-B1	UperNet	60.255M	140.897

models. Considering that models with larger Params often have stronger fitting capabilities, to ensure the fairness of the experiment, we scaled all the models to a uniform Params size of ∼60 M to train them from scratch. The scaling rule does not alter the network's paradigm; it mainly adjusts the depth and width of the network. For instance, the original NLFNet employs ResNet-18 as its backbone; we replaced the backbone with ResNet-34 and adjusted the decoder's width (the number of channels in the decoder feature map) accordingly to approximate the scale of ∼60 M. For an input size of $3 \times 512 \times 512$, the details of all the selected models are provided in Table I. Among them, the first nine are monomodal models, while the last six are bimodal models.

C. Implementation Details

The experimental hardware used was an NVIDIA GeForce RTX 4090 GPU with 24-GB memory. To achieve rapid convergence, we utilized the AdamW optimizer with momentum $\{\beta 1 = 0.9,\ \beta 2 = 0.999\}$, and to prevent overfitting, the L2 weight decay was set to 0.02. We employed the Apex training trick with an O0 optimization level. The batch size is set to the maximum value allowed by the available GPU's memory. All the models underwent training for 100 epochs using a single-precision floating point. A linear learning rate warm-up was applied for the first five epochs, followed by a cosine annealing strategy to adjust the learning rate. The change in the learning rate is given as follows:

$$\mathrm{lr} = \begin{cases} \left[0.2375\ \mathrm{epoch} + 0.02\right] \cdot \mathrm{lr}_0, & \mathrm{epoch} \le 5 \\ \left[0.475\left(1+\cos\left(\dfrac{\mathrm{epoch}-4}{95}\right)\right) + 0.02\right] \cdot \mathrm{lr}_0, & \mathrm{else} \end{cases} \tag{24}$$

where the constant terms are empirical and result-driven, and lr_0 denotes the initial learning rate of each model. Each model fine-tunes hyperparameters and retains the best results. We normalized the input data by using the values located at "3σ" of the normal distribution for each modality as the denominator. Since data augmentation becomes more complex when dealing with modality-agnostic scenarios [63], we only employed random resize cropping at ratios ranging from 0.75 to 1.5.

The evaluation metrics used in this study included two major categories. The first category was to evaluate the accuracy of the network, including the $F1$-score, intersection over union (IoU), overall accuracy (OA), and mean IoU (mIoU). The second category was to evaluate the network's efficiency, including Params, FLOPs, and frames per second (FPS), where FPS indicates the amount of data the network processes per second and serves as a measure of the network's speed.

V. BENCHMARK EXPERIMENTAL RESULTS

A. Results on the ISPRS Potsdam Dataset

We use the official accuracy evaluation metrics of the Potsdam benchmark, per-class $F1$-score, and OA to assess the accuracy of each model. Considering that RGB imagery is the most commonly used data source in monomodal semantic segmentation, we choose this data type as the input. For the bimodal benchmark, after fully assessing the distinctiveness of the other data sources in comparison to that of RGB imagery, the DSM is selected as the second modality input. Since our MultiSenseSeg has the ability to process more modalities, we also evaluate its performance in such cases. All the results are reported in Table II, where *MultiSenseSeg-n* denotes MultiSenseSeg with n input modalities. The visual results are presented in Fig. 9.

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TABLE II
ACCURACY RESULTS ON THE ISPRS POTSDAM DATASET (%)

Model	Modal	Impervious surface	Building	Low vegetation	Tree	Car	OA
TransUNet	RGB	83.179	89.858	76.901	80.799	83.580	85.152
MANet	RGB	83.446	90.018	77.775	80.865	84.940	85.336
UNetFormer	RGB	83.129	89.786	77.450	79.846	85.313	84.834
CMT	RGB	83.643	90.052	78.090	80.222	84.384	85.228
Swin Transformerv2	RGB	82.728	89.395	77.031	78.656	83.311	84.345
ConvNeXtv2	RGB	81.849	88.610	75.315	76.516	83.303	83.352
RMT	RGB	83.374	89.745	77.949	80.731	84.931	85.149
SwinUNet	RGB	81.564	88.351	75.317	76.338	80.815	82.997
EGENet	RGB	83.473	90.365	78.297	81.417	85.171	85.564
EAFNet	RGB+DSM	84.124	91.769	78.211	81.383	84.264	86.105
NLFNet	RGB+DSM	84.110	91.670	78.302	81.150	85.103	86.085
UisNet	RGB+DSM	78.103	86.628	69.529	68.483	4.253	79.309
DIR_DeepLabv3+	RGB+DSM	84.006	91.510	78.269	81.304	84.670	86.027
D_UNet	RGB+DSM	84.114	91.292	77.742	81.864	85.409	86.007
CMX	RGB+DSM	83.687	91.480	78.244	80.690	83.736	85.847
MultiSenseSeg-1	RGB	83.939	90.352	78.859	81.789	84.747	85.881
MultiSenseSeg-2	RGB+DSM	84.713	91.842	79.292	82.585	85.506	86.768
MultiSenseSeg-3	RGB+DSM+IRRG	84.782	**91.858**	**79.878**	**82.915**	**85.606**	**87.131**
MultiSenseSeg-4	RGB+DSM+IRRG+RGBIR	**84.984**	91.852	79.803	82.713	85.515	86.973

1) Comparison With the SOTA Models: The results indicate that at similar model scales, whether in OA or per-class $F1$-score, the accuracy of MultiSenseSeg consistently surpasses that of the previous models. Our method can achieve SOTA performance under homogeneous input modal types and allows for further performance gains with extra available modalities. Specifically, when the input modality is consistent, i.e., in the monomodal benchmark, MultiSenseSeg-1, which uses only RGB data, outperforms all the monomodal models in the control group. Similarly, in the bimodal benchmark, the accuracy of MultiSenseSeg-2 also surpasses the best result in the bimodal control group. Furthermore, when MultiSenseSeg leverages its advantage in accommodating more modalities, it further pushes the performance boundaries.

2) Analysis of Using Different Modalities: The results show that both DSM and IRRG carry beneficial segmentation features, as the dual-input and triple-input performances of MultiSenseSeg consistently outperform that of the single-modality representation. Moreover, the explainability of modality advantages also meets expectations. For instance, the "building" class, which is most sensitive to elevation information, achieves a significant improvement of ∼1.5% upon incorporating DSM. As shown in Fig. 9, the marked building parts, due to the confusing RGB information, are highly susceptible to false negative misclassification, and this phenomenon is significantly improved by the assistance of DSM in all the shown multimodal models. The quantitative results of MultiSenseSeg-3 and MultiSenseSeg-4 show that merely altering the channel combination to stack homogeneous feature bases cannot introduce new visual representations to the network and may even lead to slight degradation due to an increase in redundant information. In addition, the improvement from extrapolating from MultiSenseSeg-1 to MultiSenseSeg-2 is much greater than that from MultiSenseSeg-2 to MultiSenseSeg-3 because the extra modality feature added in "RGB + DSM + IRRG" compared to "RGB + DSM" is actually just the "IR" band. The redundant "R&G" bands might excessively emphasize homogeneous features, increasing the burden on the MSEs and AMM module of MultiSenseSeg. These findings indicate that the performance gains brought about by multimodality to the network indeed come from the inherent possession of different perceptual dimensions by the modalities themselves rather than from the simple combination of feature reuse and model extrapolation.

B. Results on the ZJU-RGB-P Dataset

For the ZJU-RGB-P dataset, the official accuracy metrics authors used are per-class IoU and mIoU [52]. Like in the Postdam benchmark, RGB remains the monomodal input, while, for the bimodal setup, we use the DoLP as the second modality input. In addition, we also report the results of MultiSenseSeg with inputs of three, five, and nine modalities. The quantitative results are reported in Table III, and visual results are presented in Fig. 10.

1) Comparison With the SOTA Models: In both the monomodal and bimodal benchmarks of the ZJU-RGB-P dataset, MultiSenseSeg's mIoU significantly outperforms that of the control group models. The optimal segmentation accuracy for all the other classes comes from different versions

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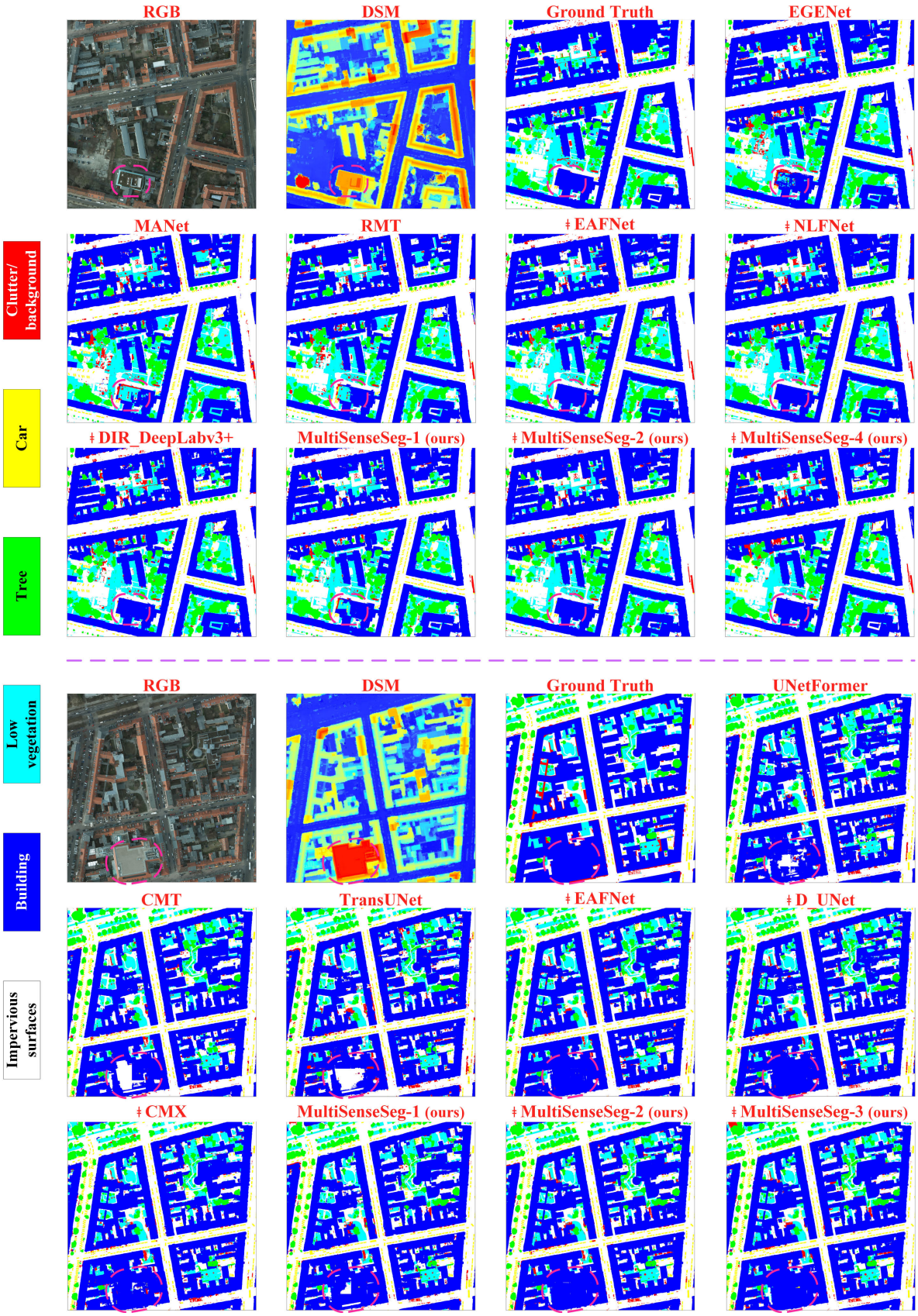


Fig. 9. Visualization comparisons on the ISPRS Potsdam dataset, where "‡" indicates multimodal models.

of MultiSenseSeg. Especially when dealing with the most challenging "pedestrian" class, MultiSenseSeg-1, without the use of polarization data assistance, outperforms its strongest competitor, NLFNet, by ∼2.2% of the IoU. After incorporating

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TABLE III
ACCURACY RESULTS ON THE ZJU-RGB-P DATASET (%), WHERE L_1 DENOTES $I_{0^\circ} + I_{45^\circ} + I_{90^\circ} + I_{135^\circ}$; L_2 DENOTES RGB(S_0) + S_1 + S_2; AND L_3 DENOTES DOLP + AOP

Model	Modal	Building	Glass	Car	Road	Vegetation	Sky	Pedestrian	Bicycle	mIoU
TransUNet	RGB	85.522	79.301	93.660	96.340	92.222	85.718	43.643	76.048	81.557
MANet	RGB	85.449	78.434	93.459	96.428	92.321	84.751	50.519	81.183	82.818
UNetFormer	RGB	85.560	78.219	92.649	96.500	92.653	84.651	48.152	83.802	82.773
CMT	RGB	88.034	80.892	94.117	96.573	93.204	86.756	54.666	86.646	85.111
Swin Transformerv2	RGB	87.383	73.177	92.218	95.915	92.567	86.928	51.809	85.774	83.222
ConvNeXtv2	RGB	84.907	79.063	92.399	95.910	91.988	84.957	44.589	78.727	81.568
RMT	RGB	87.232	79.294	93.994	96.380	92.731	87.111	42.358	84.548	82.956
SwinUNet	RGB	68.529	52.303	81.500	91.545	89.132	75.450	12.029	68.601	67.386
EGENet	RGB	87.272	80.296	93.994	96.579	92.958	85.017	50.910	84.742	83.971
EAFNet	RGB+DoLP	83.895	74.930	91.884	96.107	92.622	86.593	55.412	83.268	83.089
NLFNet	RGB+DoLP	86.552	80.666	94.288	96.508	93.120	86.151	60.787	81.558	84.954
UisNet	RGB+DoLP	50.623	19.850	68.332	83.533	84.992	59.430	5.693	49.031	52.685
DIR_DeepLabv3+	RGB+DoLP	85.387	76.912	93.087	96.189	92.812	84.853	55.826	82.951	83.502
D_UNet	RGB+DoLP	81.653	75.725	90.314	95.434	91.424	81.528	33.051	72.584	77.714
CMX	RGB+DoLP	86.241	78.655	93.501	96.426	92.833	85.647	54.974	84.097	84.047
MultiSenseSeg-1	RGB	86.743	79.553	93.861	**96.627**	93.220	86.096	62.975	86.537	85.702
MultiSenseSeg-2	RGB+DoLP	87.341	**82.110**	94.096	96.338	**93.457**	**87.200**	67.476	86.862	86.860
MultiSenseSeg-3	RGB+L_3	85.241	81.973	**94.584**	96.574	93.169	86.075	59.256	**87.103**	85.497
MultiSenseSeg-5	L_2+L_3	**87.396**	80.254	94.240	96.596	93.303	87.118	64.262	86.800	86.246
MultiSenseSeg-9	L_1+L_2+L_3	86.305	81.689	94.571	96.501	93.307	85.302	**71.068**	86.347	**86.886**

the DoLP, the gap further widened to ∼7.5%. The proposed novel paradigm can not only achieve competitive accuracy in bird's-eye view remote sensing data but is also suitable for natural scene ground-based photogrammetric sensor data.

2) Analysis of Using Different Modalities: For the ZJU-RGB-P dataset, an interesting phenomenon is that the assistance of polarization information does not always translate to an improvement in accuracy. When using the RGB-DoLP input form, the overall performance of MultiSenseSeg significantly improved, but, with the addition of AoP data, the accuracy started to decline and became even worse than that when using only RGB imagery. As more modalities are incorporated, the accuracy gradually begins to increase and reaches its optimum. This phenomenon may occur because too many abstract polarization state representations overshadow the importance of RGB information in the feature space until more data containing spatial details are added to guide accurate segmentation. In addition, different combinations of modalities do not exhibit a consistent pattern in their contributions to each subclass. For instance, the input form of nine modal combinations is optimal for segmenting the "pedestrian" class, while the input form that adds only DoLP improves the segmentation of "glass," "vegetation," and "sky" most significantly. This highlights the importance of carefully selecting multimodal data types based on specific tasks (e.g., introducing all nine modalities is a wise choice for pedestrian detection tasks but not for vegetation extraction). On the other hand, despite the presence of jitter, the addition of polarization information is always beneficial for segmenting the "glass" and "car" classes. This is because polarization data inherently suit objects such as glass, which exhibit surface reflectivity or are highly sensitive to the propagation of light. This is consistent with the performance of MultiSenseSeg and demonstrates its paradigm adaptability.

3) Stability Analysis of Multimodal Training: Training for more modalities is challenging because it may accompany gradient disorder. As shown in Fig. 11, we provide the changes in loss and mIoU during the training process of MultiSenseSeg. The figure demonstrates that even when facing the case of simultaneously inputting all nine modalities, the curve changes smoothly without any NaN values, indicating robustness. The proposed method can easily achieve end-to-end training in multimodal cases.

C. Computational Cost Increment of Model Extrapolation

One of the topics of this study is to reduce the extrapolation cost. Hence, in Table IV, we report the incremental changes in Params, FLOPs, FPS, and tensor memory for MultiSenseSeg when an extra input modality is added under multiple scales. Pseudosamples based on the standard Gaussian distribution are used for testing, with three different scales: 256 × 256, 512 × 512, and 1024 × 1024, each with three channels. As a reference, we extended the bimodal models with extrapolation capability according to their feature fusion paradigms to further accommodate more modalities. Since the extrapolation process requires manual rewriting of the original network structure, considering the redesign burden, we utilized two models that possess extrapolation capability, DIR_DeepLabv3+ and D_UNet, and extrapolated them to the case of adding just one extra modality based on their paradigms (as shown in Table V). All the results were obtained

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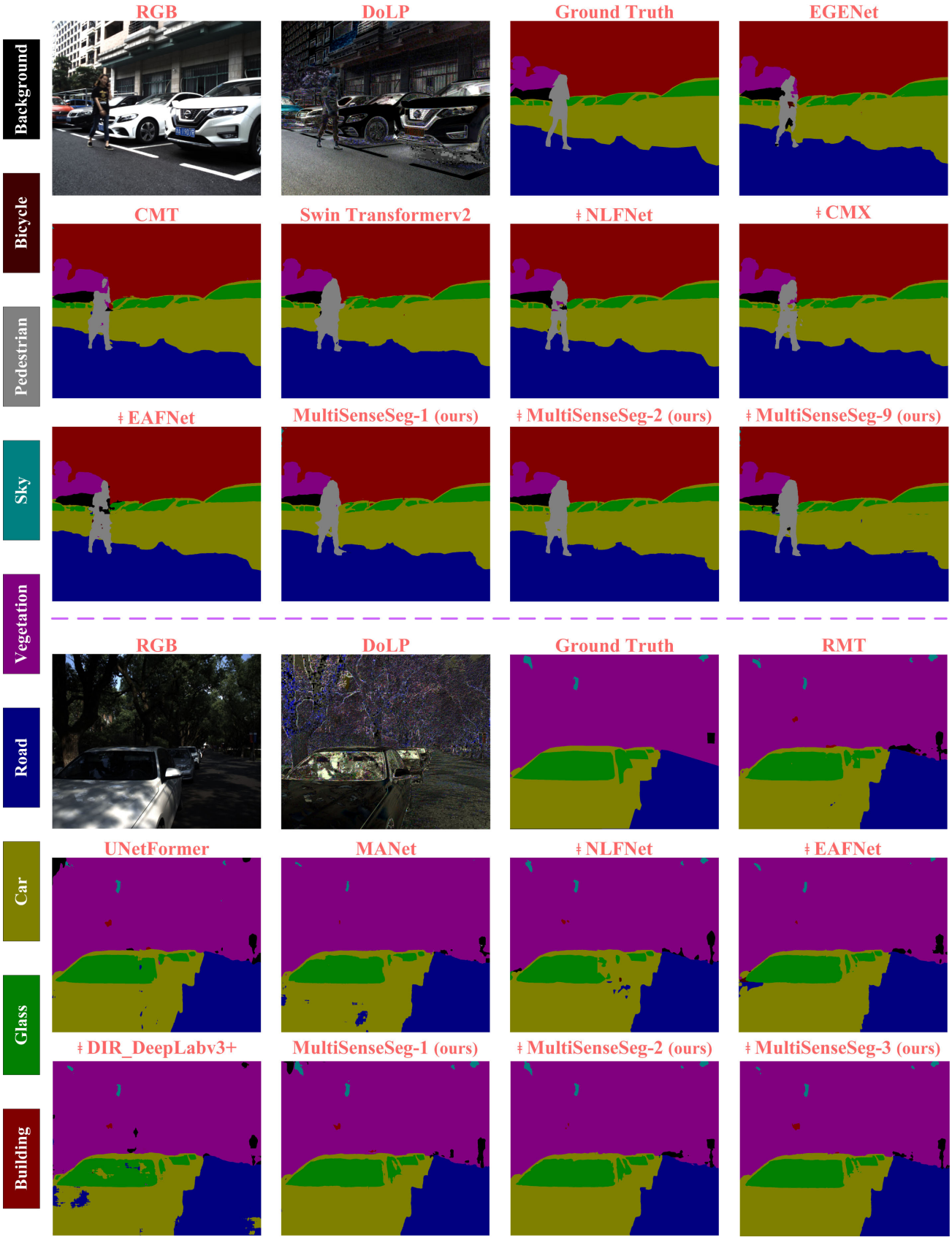


Fig. 10. Visualization comparisons on the ZJU-RGB-P dataset, where "‡" indicates multimodal models.

using an RTX 4090 GPU with the CentOS 7.9 operating system (Torch 2.0.1 + Cuda 11.8).

The results demonstrate that the computational cost associated with adding an extra modality to our MultiSenseSeg is

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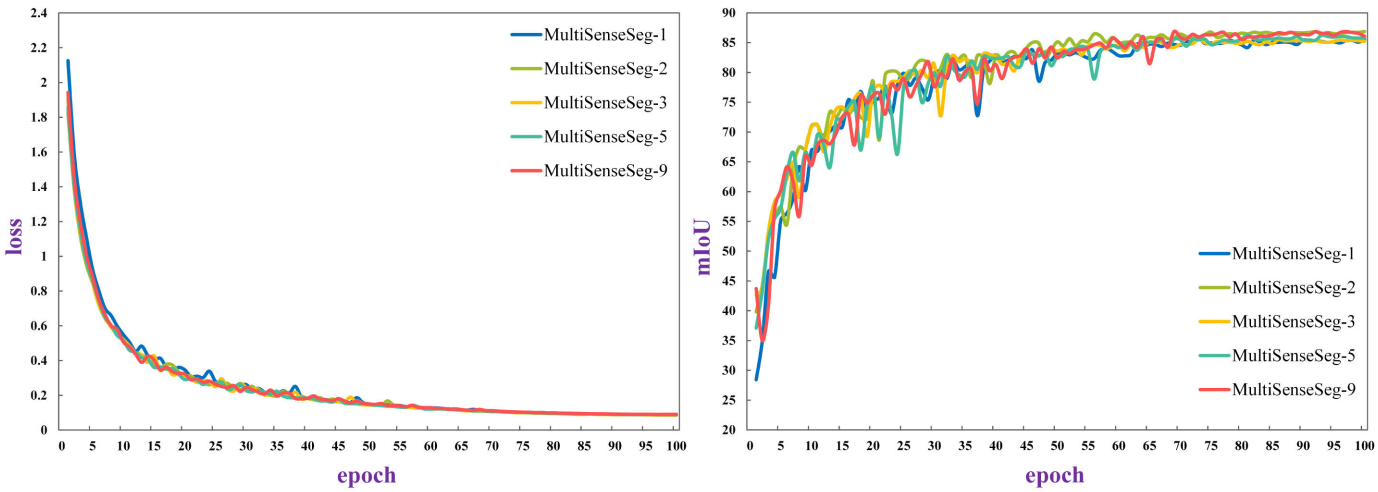


Fig. 11. Changes in loss (left) and mIoU (right) of MultiSenseSeg during training on the ZJU-RGB-P dataset.

TABLE IV
COMPUTATIONAL COST INCREMENTS OF MULTISENSESEG

Image scale	Modal quantity	Params	FLOPs	FPS	Memory
(3, 256, 256)	1	58.5861M	65.0890G	54.2858	234.8774M
	2	58.6585M	66.3379G	51.1508	235.9194M
	3	58.7330M	67.5963G	49.3964	236.9775M
	4	58.8095M	68.8646G	47.6670	238.0508M
	10	59.3117M	76.7101G	40.7632	246.0796M
	20	60.3125M	90.9066G	32.1695	258.2930M
(3, 512, 512)	1	58.5861M	260.2963G	40.5339	238.5493M
	2	58.6585M	265.2904G	38.3614	242.2163M
	3	58.7330M	270.3190G	36.6899	244.4307M
	4	58.8095M	275.3828G	35.2637	248.4727M
	10	59.3117M	306.5295G	28.7313	268.8765M
	20	60.3125M	361.5745G	21.1843	305.1992M
(3, 1024, 1024)	1	58.5861M	1.0411T	17.6645	246.1274M
	2	58.6585M	1.0611T	15.5290	258.4194M
	3	58.7330M	1.0812T	14.0059	270.7275M
	4	58.8095M	1.1015T	12.7858	283.6699M
	10	59.3117M	1.2258T	8.0682	358.5796M
	20	60.3125M	1.4442T	4.8089	483.2930M

extremely minimal. In contrast, the cost of adding one additional input to the reference models is substantial. Specifically, adding one extra input in MultiSenseSeg increases the Params by just ∼0.08 M Params, whereas, for DIR_DeepLabv3+ and D_UNet, they require ∼26 and ∼20 M of Params, respectively, which are 250 and 325 times higher than those of our method, respectively. Moreover, MultiSenseSeg maintains robustness in terms of computational complexity (FLOPs) and temporal complexity (FPS), while the comparison models experience a sharp increase in computational load, and the inference speed is also significantly slowed. As for memory, without considering cache, the MultiSenseSeg's increased tensor memory for each extra modality can be approximated by storing the Params of a lightweight MSE and its corresponding feature maps. In comparison to adding a complete backbone, it is evident that the former has better extrapolation properties. Furthermore, our method exhibits more obvious advantages when processing high-resolution data, as larger feature tensors occupy more memory in the extrapolation part of the model. At a resolution of 1024 × 1024, our MultiSenseSeg requires ∼12 M memory for storing extra modality, while the two comparison methods increased by ∼113 and ∼87 M, respectively.

It is clear that traditional feature fusion paradigms have shown sensitivity to the number of inputs, demanding

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TABLE V
COMPUTATIONAL COST INCREMENTS OF DIR_DEEPLABV3+ AND D_UNET

Image scale	Model	Modal quantity	Params	FLOPs	FPS	Memory
(3, 256, 256)	DIR_DeepLabv3+	2	60.4613M	41.1225G	58.4798	237.2129M
		3	86.0788M	58.0555G	40.2655	338.6714M
	D_UNet	2	60.3673M	199.9169G	136.5843	233.7549M
		3	80.1202M	263.1734G	103.9412	309.9268M
(3, 512, 512)	DIR_DeepLabv3+	2	60.4613M	164.4893G	53.4124	239.9629M
		3	86.0788M	232.2210G	32.3828	344.4214M
	D_UNet	2	60.3673M	0.7997T	31.6458	238.3066M
		3	80.1202M	1.0527T	24.1574	316.6772M
(3, 1024, 1024)	DIR_DeepLabv3+	2	60.4613M	657.9565G	27.5530	259.7129M
		3	86.0788M	928.8834G	18.8943	372.4214M
	D_UNet	2	60.3673M	3.1987T	6.9723	256.2549M
		3	80.1202M	4.2108T	5.3421	343.6768M

TABLE VI
ABLATION RESULTS WHEN USING DIFFERENT FEATURE EXTRACTION PIPELINES AND DECODER NECKS

Pipeline	Decoder neck	Accuracy		Params (M)	GFLOPs	FPS
		ISPRS Potsdam (%)	ZJU-RGB-P (%)			
ConvNeXtv2	UperNet	85.435	81.542	61.722	267.197	54.963
Swin Transformerv2		86.380	84.351	61.433	261.341	39.197
CMT		86.327	85.526	63.581	261.412	30.692
RMT		86.262	83.882	56.834	264.095	30.395
MultiSenseSeg-2	FPN	86.640	86.262	44.499	262.049	39.241
	UNet	86.626	84.047	34.460	53.932	50.593
	UNetFormer	86.231	84.584	55.486	162.393	40.632
MultiSenseSeg-2	UperNet	86.768	86.860	58.659	265.290	38.361

substantial costs that restrict their applicability to broader scenarios. In contrast, MultiSenseSeg's novel scaling-friendly structure remains highly competitive, even in the extreme scenario of inputting 20 different modalities simultaneously. These demonstrate the feasibility of the proposed method for addressing quantity-agnostic segmentation tasks.

VI. ABLATION STUDY

To explore how different factors affect the performance of MultiSenseSeg, we performed a series of ablation studies. We first conduct ablation experiments on two indispensable parts, the feature extraction pipeline and the decoder neck (UperNet) separately, and then evaluate the effectiveness of the remaining components. Subsequently, we conduct experiments on a key technique in multimodal semantic segmentation: data augmentation. Finally, we analyze the main hyperparameters of our model. Without loss of generality, the models' input form is taken as bimodal for example (i.e., RGB + DSM for the ISPRS Potsdam dataset and RGB + DoLP for the ZJU-RGB-P dataset).

A. Feature Extraction Pipeline and Decoder Neck

We evaluate the performance of the proposed method by selecting potential alternative solutions for these two parts from the SOTA or popular methods. Specifically, for the feature extraction pipeline, to ensure that the selected approaches fully cover operator characteristics, we chose: 1) *ConvNeXtv2*, a representative pure convolution-based solution from the CNN camp; 2) *Swin Transformerv2*, a representative pure transformer-based solution from the ViT camp; 3) *CMT*, a representative work of the hybrid approach combining CNN and ViT; and 4) *RMT*, the visual solution of RetNet [64], a recent upsurge hailed as a promising successor to transformers. For the decoder neck, we select FPN, UNet, and UNetFormer, where the first two are popular methods and UNetFormer is an SOTA method from the remote sensing community. For objective evaluation, we scaled the feature extraction pipelines to a similar network size (∼60 M). Since the width of the decoder neck needs to be synchronized with the output channels of our multimodal backbone, our invariance for the decoder neck is set according to the width of the backbone. All results are illustrated in Table VI.

1) Analysis of Using Different Feature Extraction Pipelines: The results show that our method enables the entire network closer to the performance upper bound. The design that combines two types of operators is beneficial for multimodal segmentation because only the CMT, similar to our method based on hybrid operators, achieved relatively impressive

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TABLE VII
ABLATION RESULTS FOR DIFFERENT MULTISENSESEG VERSIONS

Version	Accuracy		Params (M)	GFLOPs	FPS
	ISPRS Potsdam (%)	ZJU-RGB-P (%)			
faster MultiSenseSeg-2	86.386	85.223	29.963	77.375	113.605
normal MultiSenseSeg-2	86.768	86.860	58.659	265.290	38.361

performance on both datasets. ConvNeXtv2 performs the worst because it utilizes dense large convolutional kernels to capture local neighborhood relationships while obtaining a large receptive field. The learned connection weights from this design are static and introduce strict inductive bias, which prevents the network from learning stronger and more robust patterns from the rich multimodal information. In contrast, benefiting from flexible adaptive spatial aggregation and sparse connections, network dynamics and regularization capabilities are enhanced, and the remaining methods show stronger adaptability to multimodal features. Moreover, at a similar network scale, ConvNeXtv2 has faster inference speeds; i.e., its pure convolutional structure possesses better parallelism and hardware support on GPU devices. Through careful design, our method can achieve higher accuracy while maintaining comparable or slightly faster speeds than the other three SOTA methods.

2) Analysis of Using Different Decoder Necks: For the three alternative solutions, the FPN achieves accuracy close to that of UperNet. This indicates that the remaining parts of MultiSenseSeg already possess some ability to extract global contextual semantics, as UperNet is composed of FPN and PPM, where the purpose of PPM is precisely to integrate contextual features. Removing PPM can save ∼14 M significant Params. However, as this operation is based on low-resolution feature maps, its impact on computational complexity (GFLOPs) and speed (FPS) is not particularly evident. Although the other two methods exhibit less accuracy, their GFLOPs are significantly lighter. These also reveal that our model has the potential for further lightweighting. Notably, since the invariant of the decoder neck part is set to our backbone width, according to the GFLOPs results of the UNet method, it can be inferred that out of the total 265.29 GFLOPs of the complete MultiSenseSeg-2, less than 1/5 comes from our core contribution—the multimodal backbone (the measured Params and GFLOPs of the backbone part are only 24.941 M and 31.292, respectively). As UperNet does not belong to this study's contribution and we merely adopt an SOTA solution from monomodal tasks, we do not further comment on it here. Employing different decoding structures could be a good choice for users with varying requirements to balance accuracy and computational costs.

3) Analysis of Using Lighter Designs for Feature Extraction Pipelines and Decoder Necks: We further relax the original high-order feature extraction approach of MultiSenseSeg and introduce a more lightweight variant to experiment with the application potential of MultiSenseSeg's paradigm for users with higher speed requirements. Inspired by the above results, we delete the self-attention operator in the feature extraction basic block (as self-attention usually comes with high computational complexity and low speed), replace it with pure convolution with better parallel capability, and merge the two residual structures into one. Considering the overfitting risk associated with densely large kernels, we abandoned the directly large receptive field; at this point, the basic block was degraded to a standard regular convolution residual block with small kernel static weights only (like the ResNet block, which contains only first-order operations). Meanwhile, we correspondingly compress the overall network scale, reducing its Params to approximately half of the original scale. In Table VII, we report the performance of the modified variant of MultiSenseSeg following the above alterations, where "faster" denotes the variant version and "normal" represents the original MultiSenseSeg.

Based on the aforementioned designs, our model's inference speed has increased by approximately three times, and the computational complexity has significantly decreased. However, there is an unavoidable reduction in accuracy. This aligns with our expectations, allowing for some tradeoffs in accuracy to gain faster speed. In addition, compared to the ZJU-RGB-P dataset, the variant version experiences only a minimal decrease in accuracy on the Potsdam benchmark. This phenomenon could be explained by the differences in the datasets, i.e., data sources and content nature. Specifically, the Potsdam dataset is captured by satellite sensors, while ZJU-RGB-P comprises side-view data from natural scenes. As spaceborne sensors capture objects from distant spaces, the captured objects are small and densely packed, requiring finer local details for differentiation. In such cases, using a pure CNN structure can already extract satisfactory semantics. However, the sensors used for oblique photogrammetry are relatively close-range and capture large continuous features that require larger receptive fields in networks. Nevertheless, increasing the receptive field in CNNs relies on deeper networks, and even very deep CNNs cannot match the theoretical receptive field in practice. Removing the self-attention operator, which has a large effective receptive field, directly results in a significant decrease in accuracy for the ZJU-RGB-P dataset. Therefore, for natural scenes or close-range oblique unmanned aerial vehicle perspective (e.g., UAVid) datasets, we recommend introducing the self-attention operator. In addition, looking back at the results from Tables II and III, even with our lightweight "faster" version of MultiSenseSeg, it outperforms all models in the control group. This further reveals the effectiveness of our method.

B. Remaining Components of MultiSenseSeg

1) Necessity of Each Component: To demonstrate the contribution of each remaining component, we constructed the following sets.

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TABLE VIII
ABLATION STUDY OF EACH COMPONENT OF MULTISENSESEG (%)

Baseline	MSEs+AMM	DG	Auxiliary classifier	ISPRS Potsdam	ZJU-RGB-P
√	-	-	-	85.586	83.723
√	√	-	-	$86.415_{+0.829}$	$84.467_{+0.744}$
√	-	√	-	$85.962_{+0.376}$	$83.807_{+0.084}$
√	-	-	√	$86.458_{+0.872}$	$84.832_{+1.109}$
√	√	√	-	$86.293_{+0.707}$	$85.105_{+1.382}$
√	√	-	√	$86.595_{+1.009}$	$85.368_{+1.645}$
√	-	√	√	$86.434_{+0.848}$	$84.510_{+0.787}$
√	√	√	√	$86.768_{+1.182}$	$86.860_{+3.137}$

1) *Baseline:* Coupling of two indispensable parts—the feature extraction pipelines and the decoder neck—while adding a patch embedding layer (a common layer of ViTs) to maintain model logical consistency.
2) *MSEs + AMM:* Since the input of the AMM module is the high-order feature maps processed by MSEs and the outputs of MSEs require AMM low-rank processing before being fed into the subsequent feature extraction pipeline, we combined them for an ablation study.
3) *DG:* Ablation for the DG only.
4) *Auxiliary Classifier:* Ablation for the Auxiliary classifier only.

As shown in Table VIII, based on the aforementioned sets, a total of eight different experimental groups can be obtained through permutation and combination. The addition or removal of each component results in varying degrees of accuracy improvement or loss, demonstrating the necessity of each component.

2) Necessity of Independent Intramodal Extractors: In multimodal cases, the combination of MSEs + AMM can bring significant performance gains, especially on the ZJU-RGB-P dataset, where removing MSEs + AMM led to a 2.35% mIoU loss (86.86 versus 84.51). Looking back at Fig. 8, there is a significant visual distribution difference between RGB and DoLP; due to the lack of independent intramodal feature extractors available to overcome this large gap, the network cannot fully leverage diverse information sources. Another compelling argument revealing the necessity of independent intramodal extractors is given as follows: reviewing the unsatisfactory performance of UisNet in the benchmark experiments (as given in Tables II and III), we objectively speculate that the reason is that UisNet is the only model employing a multibranch Siamese fusion strategy, i.e., the weights of its two branches are shared rather than independent. To demonstrate this point more strongly, we select an extra model with a multibranch Siamese fusion strategy—LASNet [15]—for further testing. Its accuracies on the two datasets are 54.389% and 45.349%, respectively. When we input both branches with RGB data, the accuracies reach 82.477% and 71.272%, respectively. These findings indicate that when there is a large gap between two input modalities, the network's weights struggle to provide consistent decisions across modalities. In such cases, cross-modal homogenous operation becomes ineffective or even detrimental. Therefore, to design a unified model for quantity-agnostic modalities, we strongly recommend the use of independent intramodal extractors to cover modalities with significant data distribution differences.

C. Multimodal Data Augmentation

Data augmentation, a common technique in monomodal data, becomes challenging in multimodal scenarios due to modality agnosticism. Specifically, traditional data augmentation methods such as rotation and flipping, commonly applied to monomodal images, may not be entirely suitable for multimodal cases due to data characteristics. For instance, because of the polarimetric characteristics, the horizontal flip of the AoP is not a simple spatial flip but rather 180°-AoP', where AoP' denotes merely the spatially horizontal flipped version of the initial AoP data. As part of a unified algorithm, to avoid the risk of data distortion when dealing with modality agnosticism, we utilize only overlapping and random resize cropping, which are universally applicable to all modalities. Taking the ISPRS Potsdam dataset as an example, as shown in Table IX ("w/o" stands for MultiSenseSeg without any data augmentation, and "w" stands for results with the aforementioned data augmentation methods), significant accuracy improvements can be achieved by employing only a limited set of data augmentation methods compared to no augmentation at all. This indicates that for multimodal cases, suitable data augmentation techniques are still an important way to obtain performance gains. However, there is still a long way to go in exploring universal data augmentation methods, and this could become a major prospect for the future.

D. Model Hyperparameter Analysis

The model's hyperparameters determine the scale and performance of the model. To explore how different compositions of hyperparameters affect the model's performance, we introduce the main hyperparameter information of MultiSenseSeg in Table X. These hyperparameters respectively control the width and depth of the network.

According to these hyperparameters, we manually construct three different versions of MultiSenseSeg for ablation studies.

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TABLE IX
PERFORMANCE WITHOUT AND WITH DATA AUGMENTATION ON THE ISPRS POTSDAM DATASET (%)

Modal	Method	Impervious surfaces	Building	Low vegetation	Tree	Car	OA
RGB	w/o	82.637	89.116	77.161	79.035	83.353	84.269
	w	83.939	90.352	78.859	81.789	84.747	85.881
RGB+DSM	w/o	83.011	90.117	77.702	80.099	84.124	85.090
	w	84.713	91.842	79.292	82.585	85.506	86.768
RGB+DSM+IRRG	w/o	83.195	89.803	78.387	81.451	84.163	85.276
	w	84.782	91.858	79.878	82.915	85.606	87.131
RGB+DSM+IRRG+RGBIR	w/o	83.364	89.869	78.440	81.298	83.893	85.272
	w	84.984	91.852	79.803	82.713	85.515	86.973

TABLE X
HYPERPARAMETERS AND INTERPRETATION OF MULTISENSESEG

Hyperparameter	Paraphrase	Binding rules
modal dim	dimension of each modality in the MSE	can be divisible by group dim
embed dim	dimension of the 1st stage in the pipeline	can be divisible by h_i and group dim
decoder dim	channel numbers of the decoder.	N^+
depths	number of basic blocks for the pipeline	$\{A, A, B, A\}$, $B>A \in N^+$
heads	number of heads for the pipeline	$\{h_1, h_2, h_3, h_4\}$, $h_{i+1}=2h_i$
group dim	channel numbers in each group of group convolution	N^+

TABLE XI
ABLATION RESULTS FOR MULTISENSESEG USING DIFFERENT HYPERPARAMETERS

Version	Accuracy		Params (M)	GFLOPs	FPS
	ISPRS Potsdam (%)	ZJU-RGB-P (%)			
MultiSenseSeg-2-T	86.415	84.067	7.218	18.577	60.023
MultiSenseSeg-2-S	86.544	85.253	21.817	78.753	53.339
MultiSenseSeg-2-B	86.768	86.860	58.659	265.290	38.361

1) *MultiSenseSeg-2-T (Tiny):* Modal dim = 24; embed dim = 48; decoder dim = 96; depths = {2, 2, 6, 2}; heads = {2, 4, 8, 16}; and group dim = 6.
2) *MultiSenseSeg-2-S (Small):* Modal dim = 32; embed dim = 72; decoder dim = 256; depths = {2, 2, 6, 2}; heads = {3, 6, 12, 24}, and group dim = 8.
3) *MultiSenseSeg-2-B (Base):* Modal dim = 32; embed dim = 96; decoder dim = 512; depths = {2, 2, 8, 2}; heads = {3, 6, 12, 24}; and group dim = 8.

We report the performance of the three versions in Table XI. The results show that as the network becomes lighter, there is an inevitable accuracy reduction in accuracy on both datasets. This indicates that achieving higher accuracy requires a certain model scale. Notably, even the extremely lightweight version of MultiSenseSeg still exhibits a precision advantage compared to several SOTA methods with ∼60 M Params.

VII. CONCLUSION

Semantic segmentation is a crucial technology in remote sensing. However, unlike the booming development of monomodal semantic segmentation models, research on multimodal visual models remains limited. Previous studies on multimodal research have mostly considered only dual-input cases, and the potential benefits of diverse data sources have not been fully explored. To further uncover the latent value of incorporating more modalities into semantic segmentation, this work explores a cost-effective way to address the quantity-agnostic modality question and introduces a novel multimodal semantic segmentation model, MultiSenseSeg. MultiSenseSeg not only accommodates an arbitrary number of multimodal inputs but also incurs the minimal computational cost for each extra modality. A comprehensive series of benchmark experiments and ablation studies conducted on two public multimodal datasets demonstrate the effectiveness and efficiency of MultiSenseSeg, which can achieve comparable or even superior performance to several previous SOTA methods when faced with limited modal types; when users possess additional modal data, it can further seek performance gains for users at a low incremental cost. Besides, it also has the potential to be applied to other remote sensing tasks. For example, using multitemporal data and a Siamese structure, it can capture the differences between the same spatial objects at different times, thereby being applied to remote sensing change detection tasks. Our study can provide meaningful references and offer a new option for remote sensing researchers.

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Qingpeng Wang received the B.E. degree from the College of Geoscience and Surveying Engineering, China University of Mining and Technology, Beijing, China, in 2021, where he is currently pursuing the M.E. degree. He will pursue the Ph.D. degree with the Institute of Remote Sensing and Geographic Information System, School of Earth and Space Sciences, Peking University, Beijing, in 2024.

His research interests are related to deep learning, remote sensing image processing, and high-performance geo-computation.



Wei Chen received the B.S. degree in geographical information systems from Beijing Normal University, Beijing, China, in 2008, and the Ph.D. degree in photogrammetry and remote sensing from Peking University, Beijing, in 2013.

He is currently an Associate Professor with the College of Geoscience and Surveying Engineering, China University of Mining and Technology, Beijing. His major research interests include remote sensing retrieval and remote sensing applications.



Zhou Huang received the B.Sc. degree in GIS and the Ph.D. degree in cartography and GIS from Peking University, Beijing, China, in 2004 and 2009, respectively.

He is a Professor of GIScience at the Institute of Remote Sensing and Geographical Information Systems, Peking University. He serves as the Deputy Director of the Institute of Remote Sensing and GIS, Peking University, and the Deputy Director of the Engineering Research Center of Earth Observation and Navigation, Ministry of Education, Beijing. He has published more than 100 academic papers in international journals or conferences. His main research interests include big geo-data, high-performance geo-computation, distributed geographic information processing, spatial data mining, and spatial databases.

Dr. Huang has won many awards such as the 2021 Young Cheung Kong Scholar issued to China's domestic top scholars by the Ministry of Education and the Youth Talent Innovation Plan in Remote Sensing Science and Technology in 2015 funded by the Ministry of Science and Technology of China.



Hongzhao Tang received the B.S. degree in surveying and mapping engineering from Wuhan University, Wuhan, China, in 2007, the M.S. degree in photogrammetry and remote sensing from Peking University, Beijing, China, in 2010, the M.Sc. degree in ecology from Colorado State University, Fort Collins, CO, USA, in 2013, and the Ph.D. degree in geographic information system from Peking University, in 2014.

He is currently an Associate Research Fellow at the Land Satellite Remote Sensing Application Center, Ministry of Natural Resources, Beijing. His Ph.D. research focused on radiometric calibration and spectral calibration of satellite hyperspectral sensors. His major research interests include radiometric calibration and validation of satellite sensors and hyperspectral remote sensing applications.



Lan Yang received the B.E. and M.E. degrees from China University of Mining and Technology, Beijing, China, in 2020 and 2023, respectively. She is currently pursuing the Ph.D. degree with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China.

Her research interests are related to AI for remote sensing image processing.

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